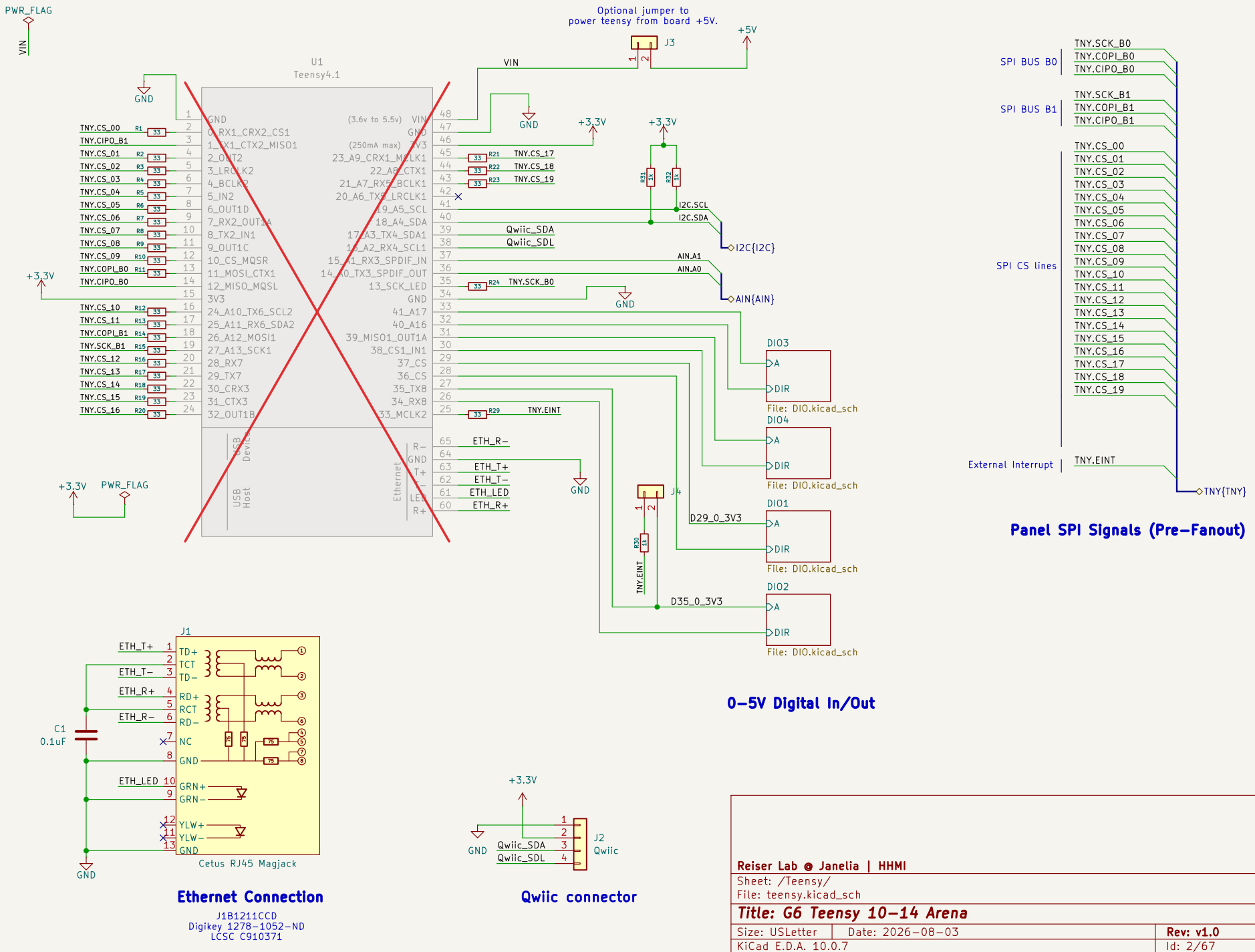


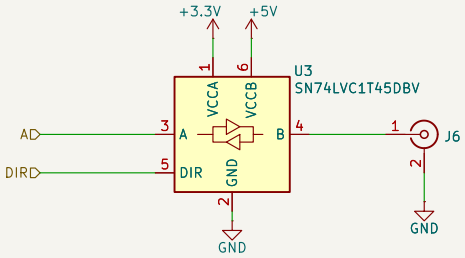
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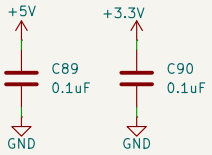
Sheet: /
File: arena_10-14.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 1/67


PWR_FLAG
VIN
Optional jumper to
power teensy from board +5V.
J3
+5V
U1
Teensy4.1
GND
+3.3V
TNY.CS_00
TNY.CIPO_B1
TNY.CS_01
TNY.CS_02
TNY.CS_03
TNY.CS_04
TNY.CS_05
TNY.CS_06
TNY.CS_07
TNY.CS_08
TNY.CS_09
TNY.COPI_B0
TNY.CIPO_B0
TNY.CS_10
TNY.CS_11
TNY.COPI_B1
TNY.SCK_B1
TNY.CS_12
TNY.CS_13
TNY.CS_14
TNY.CS_15
TNY.CS_16
TNY.CS_17
TNY.CS_18
TNY.CS_19
TNY.SCK_B0
TNY.EINT
I2C.SCL
I2C.SDA
Qwiic_SDA
Qwiic_SDL
I2C{I2C}
AIN.A1
AIN.A0
AIN{AIN}
ETH_R−
ETH_T+
ETH_T−
ETH_LED
ETH_R+
USB Device
USB Host
Ethernet
DIO3
DIO4
DIO1
DIO2
A
DIR
File: DIO.kicad_sch
J4
D29_0_3V3
D35_0_3V3
SPI BUS B0
SPI BUS B1
SPI CS lines
External Interrupt
TNY{TNY}
Panel SPI Signals (Pre−Fanout)
0−5V Digital In/Out
J1
TD+
TCT
TD−
RD+
RCT
RD−
NC
GND
GRN+
GRN−
YLW+
YLW−
C1
0.1uF
Cetus RJ45 Magjack
Ethernet Connection
J1B1211CCD
Digikey 1278−1052−ND
LCSC C910371
J2
Qwiic
Qwiic connector
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Sheet: /Teensy/
File: teensy.kicad_sch
Title: G6 Teensy 10−14 Arena
Size: USLetter
Date: 2026−08−03
Rev: v1.0
KiCad E.D.A. 10.0.7
Id: 2/67



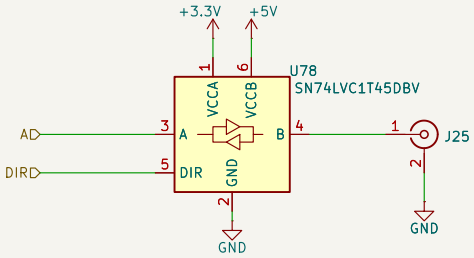


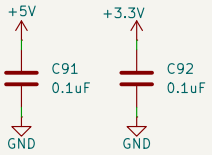
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Sheet: /Teensy/DIO2/
File: DIO.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: A4	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 49/67



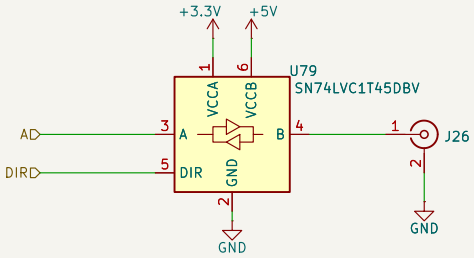


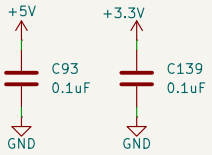
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Sheet: /Teensy/DIO1/
File: DIO.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: A4	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 50/67





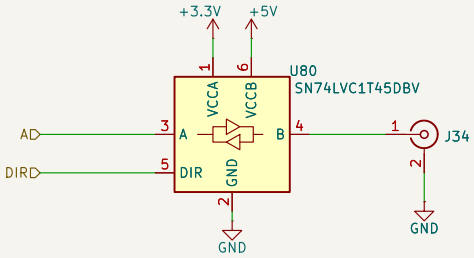
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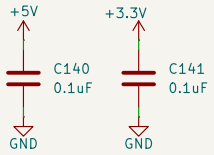
Sheet: /Teensy/DIO4/

File: DIO.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: A4	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 63/67





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Sheet: /Teensy/DIO3/
File: DIO.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: A4	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 64/67

Singals to/from Teensy

SPI BUS B0
TNY.SCK_B0
TNY.COPI_B0
TNY.CIPO_B0

SPI BUS B1
TNY.SCK_B1
TNY.COPI_B1
TNY.CIPO_B1

SPI CS lines
TNY.CS_00
TNY.CS_01
TNY.CS_02
TNY.CS_03
TNY.CS_04
TNY.CS_05
TNY.CS_06
TNY.CS_07
TNY.CS_08
TNY.CS_09
TNY.CS_10
TNY.CS_11
TNY.CS_12
TNY.CS_13
TNY.CS_14
TNY.CS_15
TNY.CS_16
TNY.CS_17
TNY.CS_18
TNY.CS_19

External Interrupt
TNY.EINT

TNY{TNY}

Pre-fanout signals

SCK B0 fan out to P1-P6

TNY.SCK_B0 → IN
OUT_1D PAN.SCK_B0_P1
OUT_2D PAN.SCK_B0_P2
OUT_3D PAN.SCK_B0_P3
OUT_4D PAN.SCK_B0_P4
OUT_5D PAN.SCK_B0_P5
OUT_6D

File: fan_out_by_5x.kicad_sch

COPI B0 fan out to P1-P6

TNY.COPI_B0 → IN
OUT_1D PAN.COPI_B0_P1
OUT_2D PAN.COPI_B0_P2
OUT_3D PAN.COPI_B0_P3
OUT_4D PAN.COPI_B0_P4
OUT_5D PAN.COPI_B0_P5
OUT_6D

File: fan_out_by_5x.kicad_sch

SCK B1 fan out to P6-P10

TNY.SCK_B1 → IN
OUT_1D PAN.SCK_B1_P6
OUT_2D PAN.SCK_B1_P7
OUT_3D PAN.SCK_B1_P8
OUT_4D PAN.SCK_B1_P9
OUT_5D PAN.SCK_B1_P10
OUT_6D

File: fan_out_by_5x.kicad_sch

COPI B1 fan out to P6-P10

TNY.COPI_B1 → IN
OUT_1D PAN.COPI_B1_P6
OUT_2D PAN.COPI_B1_P7
OUT_3D PAN.COPI_B1_P8
OUT_4D PAN.COPI_B1_P9
OUT_5D PAN.COPI_B1_P10
OUT_6D

File: fan_out_by_5x.kicad_sch

EINT fan out to P1-P5

TNY.EINT → IN
OUT_1D PAN.EINT_P1
OUT_2D PAN.EINT_P2
OUT_3D PAN.EINT_P3
OUT_4D PAN.EINT_P4
OUT_5D PAN.EINT_P5
OUT_6D

File: fan_out_by_5x.kicad_sch

EINT fan out to P6-P10

IN
OUT_1D PAN.EINT_P6
OUT_2D PAN.EINT_P7
OUT_3D PAN.EINT_P8
OUT_4D PAN.EINT_P9
OUT_5D PAN.EINT_P10
OUT_6D

File: fan_out_by_5x.kicad_sch

CS_00 fan out to P1, P6

TNY.CS_00 → IN
OUT_1D PAN.CS_00_P6
OUT_2D PAN.CS_00_P1

File: fan_out_by_2x.kicad_sch

CS_01 fan out to P1, P6

TNY.CS_01 → IN
OUT_1D PAN.CS_01_P6
OUT_2D PAN.CS_01_P1

File: fan_out_by_2x.kicad_sch

CS_02 fan out to P1, P6

TNY.CS_02 → IN
OUT_1D PAN.CS_02_P6
OUT_2D PAN.CS_02_P1

File: fan_out_by_2x.kicad_sch

CS_03 fan out to P1, P6

TNY.CS_03 → IN
OUT_1D PAN.CS_03_P6
OUT_2D PAN.CS_03_P1

File: fan_out_by_2x.kicad_sch

CS_04 fan out to P2, P7

TNY.CS_04 → IN
OUT_1D PAN.CS_04_P7
OUT_2D PAN.CS_04_P2

File: fan_out_by_2x.kicad_sch

CS_05 fan out to P2, P7

TNY.CS_05 → IN
OUT_1D PAN.CS_05_P7
OUT_2D PAN.CS_05_P2

File: fan_out_by_2x.kicad_sch

CS_06 fan out to P2, P7

TNY.CS_06 → IN
OUT_1D PAN.CS_06_P7
OUT_2D PAN.CS_06_P2

File: fan_out_by_2x.kicad_sch

CS_07 fan out to P2, P7

TNY.CS_07 → IN
OUT_1D PAN.CS_07_P7
OUT_2D PAN.CS_07_P2

File: fan_out_by_2x.kicad_sch

CS_08 fan out to P3, P8

TNY.CS_08 → IN
OUT_1D PAN.CS_08_P8
OUT_2D PAN.CS_08_P3

File: fan_out_by_2x.kicad_sch

CS_09 fan out to P3, P8

TNY.CS_09 → IN
OUT_1D PAN.CS_09_P8
OUT_2D PAN.CS_09_P3

File: fan_out_by_2x.kicad_sch

CS_10 fan out to P3, P8

TNY.CS_10 → IN
OUT_1D PAN.CS_10_P8
OUT_2D PAN.CS_10_P3

File: fan_out_by_2x.kicad_sch

CS_11 fan out to P3, P8

TNY.CS_11 → IN
OUT_1D PAN.CS_11_P8
OUT_2D PAN.CS_11_P3

File: fan_out_by_2x.kicad_sch

CS_12 fan out to P4, P9

TNY.CS_12 → IN
OUT_1D PAN.CS_12_P9
OUT_2D PAN.CS_12_P4

File: fan_out_by_2x.kicad_sch

CS_13 fan out to P4, P9

TNY.CS_13 → IN
OUT_1D PAN.CS_13_P9
OUT_2D PAN.CS_13_P4

File: fan_out_by_2x.kicad_sch

CS_14 fan out to P4, P9

TNY.CS_14 → IN
OUT_1D PAN.CS_14_P9
OUT_2D PAN.CS_14_P4

File: fan_out_by_2x.kicad_sch

CS_15 fan out to P4, P9

TNY.CS_15 → IN
OUT_1D PAN.CS_15_P9
OUT_2D PAN.CS_15_P4

File: fan_out_by_2x.kicad_sch

CS_16 fan out to P5, P10

TNY.CS_16 → IN
OUT_1D PAN.CS_16_P10
OUT_2D PAN.CS_16_P5

File: fan_out_by_2x.kicad_sch

CS_17 fan out to P5, P10

TNY.CS_17 → IN
OUT_1D PAN.CS_17_P10
OUT_2D PAN.CS_17_P5

File: fan_out_by_2x.kicad_sch

CS_18 fan out to P5, P10

TNY.CS_18 → IN
OUT_1D PAN.CS_18_P10
OUT_2D PAN.CS_18_P5

File: fan_out_by_2x.kicad_sch

CS_19 fan out to P5, P10

TNY.CS_19 → IN
OUT_1D PAN.CS_19_P10
OUT_2D PAN.CS_19_P5

File: fan_out_by_2x.kicad_sch

SPI Bus B0

PAN.SCK_B0_P1
PAN.SCK_B0_P2
PAN.SCK_B0_P3
PAN.SCK_B0_P4
PAN.SCK_B0_P5

PAN.COPI_B0_P1
PAN.COPI_B0_P2
PAN.COPI_B0_P3
PAN.COPI_B0_P4
PAN.COPI_B0_P5

PAN.CIPO_B0_P1
PAN.CIPO_B0_P2
PAN.CIPO_B0_P3
PAN.CIPO_B0_P4
PAN.CIPO_B0_P5

SPI Bus B1

PAN.SCK_B1_P6
PAN.SCK_B1_P7
PAN.SCK_B1_P8
PAN.SCK_B1_P9
PAN.SCK_B1_P10

PAN.COPI_B1_P6
PAN.COPI_B1_P7
PAN.COPI_B1_P8
PAN.COPI_B1_P9
PAN.COPI_B1_P10

PAN.CIPO_B1_P6
PAN.CIPO_B1_P7
PAN.CIPO_B1_P8
PAN.CIPO_B1_P9
PAN.CIPO_B1_P10

Chip Select P1-P10

PAN.CS_00_P1
PAN.CS_01_P1
PAN.CS_02_P1
PAN.CS_03_P1

PAN.CS_04_P2
PAN.CS_05_P2
PAN.CS_06_P2
PAN.CS_07_P2

PAN.CS_08_P3
PAN.CS_09_P3
PAN.CS_10_P3
PAN.CS_11_P3

PAN.CS_12_P4
PAN.CS_13_P4
PAN.CS_14_P4
PAN.CS_15_P4

PAN.CS_16_P5
PAN.CS_17_P5
PAN.CS_18_P5
PAN.CS_19_P5

PAN.CS_00_P6
PAN.CS_01_P6
PAN.CS_02_P6
PAN.CS_03_P6

PAN.CS_04_P7
PAN.CS_05_P7
PAN.CS_06_P7
PAN.CS_07_P7

PAN.CS_08_P8
PAN.CS_09_P8
PAN.CS_10_P8
PAN.CS_11_P8

PAN.CS_12_P9
PAN.CS_13_P9
PAN.CS_14_P9
PAN.CS_15_P9

PAN.CS_16_P10
PAN.CS_17_P10
PAN.CS_18_P10
PAN.CS_19_P10

PAN.EINT_P1
PAN.EINT_P2
PAN.EINT_P3
PAN.EINT_P4
PAN.EINT_P5

PAN.EINT_P6
PAN.EINT_P7
PAN.EINT_P8
PAN.EINT_P9
PAN.EINT_P10

PAN{PAN}

Post-Fanout Signals

CIPO B0 Enable for P1

CS0 PAN.CS_00_P1
CS1 PAN.CS_01_P1
CS2 PAN.CS_02_P1
CS3 PAN.CS_03_P1
CIPO_OUT CIPO_IN PAN.CIPO_B0_P1

File: cipo_enable.kicad_sch

TNY.CIPO_B0

CIPO B0 Enable for P2

CS0 PAN.CS_04_P2
CS1 PAN.CS_05_P2
CS2 PAN.CS_06_P2
CS3 PAN.CS_07_P2
CIPO_OUT CIPO_IN PAN.CIPO_B0_P2

File: cipo_enable.kicad_sch

CIPO B0 Enable for P3

CS0 PAN.CS_08_P3
CS1 PAN.CS_09_P3
CS2 PAN.CS_10_P3
CS3 PAN.CS_11_P3
CIPO_OUT CIPO_IN PAN.CIPO_B0_P3

File: cipo_enable.kicad_sch

CIPO B0 Enable for P4

CS0 PAN.CS_12_P4
CS1 PAN.CS_13_P4
CS2 PAN.CS_14_P4
CS3 PAN.CS_15_P4
CIPO_OUT CIPO_IN PAN.CIPO_B0_P4

File: cipo_enable.kicad_sch

CIPO B0 Enable for P5

CS0 PAN.CS_16_P5
CS1 PAN.CS_17_P5
CS2 PAN.CS_18_P5
CS3 PAN.CS_19_P5
CIPO_OUT CIPO_IN PAN.CIPO_B0_P5

File: cipo_enable.kicad_sch

CIPO B1 Enable for P6

CS0 PAN.CS_00_P6
CS1 PAN.CS_01_P6
CS2 PAN.CS_02_P6
CS3 PAN.CS_03_P6
CIPO_OUT CIPO_IN PAN.CIPO_B1_P6

File: cipo_enable.kicad_sch

TNY.CIPO_B1

CIPO B1 Enable for P7

CS0 PAN.CS_04_P7
CS1 PAN.CS_05_P7
CS2 PAN.CS_06_P7
CS3 PAN.CS_07_P7
CIPO_OUT CIPO_IN PAN.CIPO_B1_P7

File: cipo_enable.kicad_sch

CIPO B1 Enable for P8

CS0 PAN.CS_08_P8
CS1 PAN.CS_09_P8
CS2 PAN.CS_10_P8
CS3 PAN.CS_11_P8
CIPO_OUT CIPO_IN PAN.CIPO_B1_P8

File: cipo_enable.kicad_sch

CIPO B1 Enable for P9

CS0 PAN.CS_12_P9
CS1 PAN.CS_13_P9
CS2 PAN.CS_14_P9
CS3 PAN.CS_15_P9
CIPO_OUT CIPO_IN PAN.CIPO_B1_P9

File: cipo_enable.kicad_sch

CIPO B1 Enable for P10

CS0 PAN.CS_16_P10
CS1 PAN.CS_17_P10
CS2 PAN.CS_18_P10
CS3 PAN.CS_19_P10
CIPO_OUT CIPO_IN PAN.CIPO_B1_P10

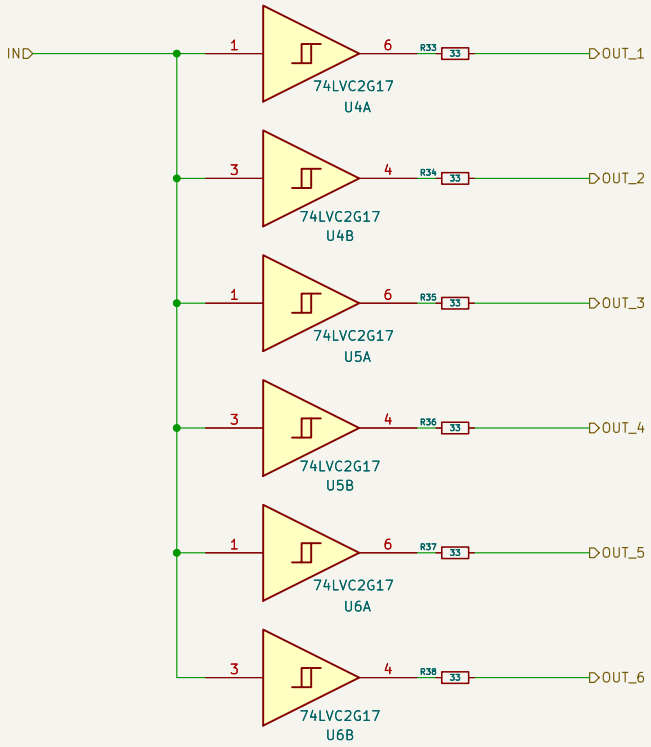
File: cipo_enable.kicad_sch

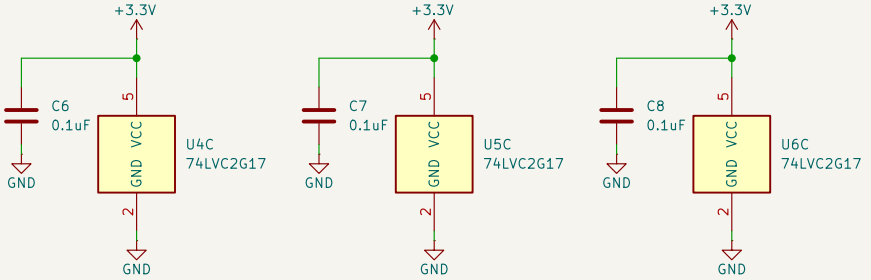
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Sheet: /Fan Out/
File: fan_out.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLedger | Date: 2026-08-03 | **Rev: v1.0**
KiCad E.D.A. 10.0.7 | Id: 3/67



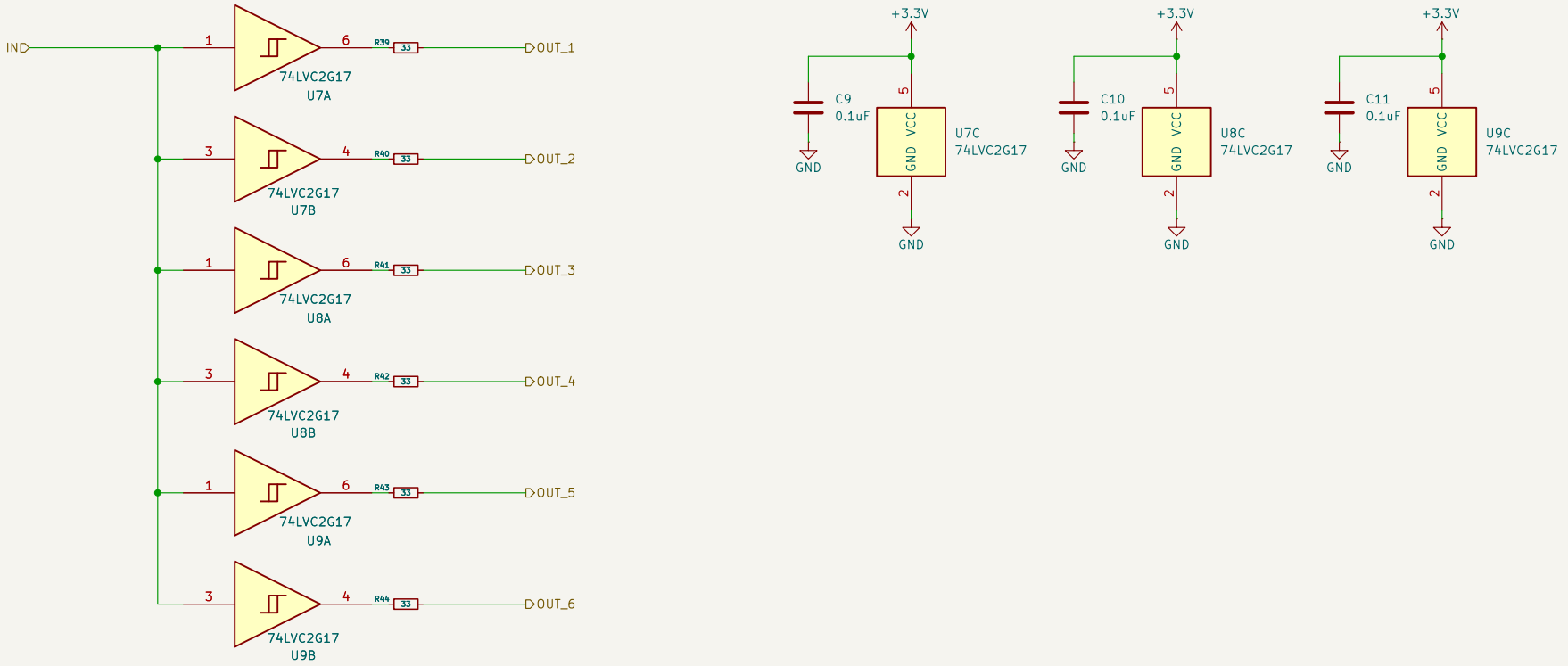


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Sheet: /Fan Out/SCK B0 fan out to P1–P6/
File: fan_out_by_5x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 4/67



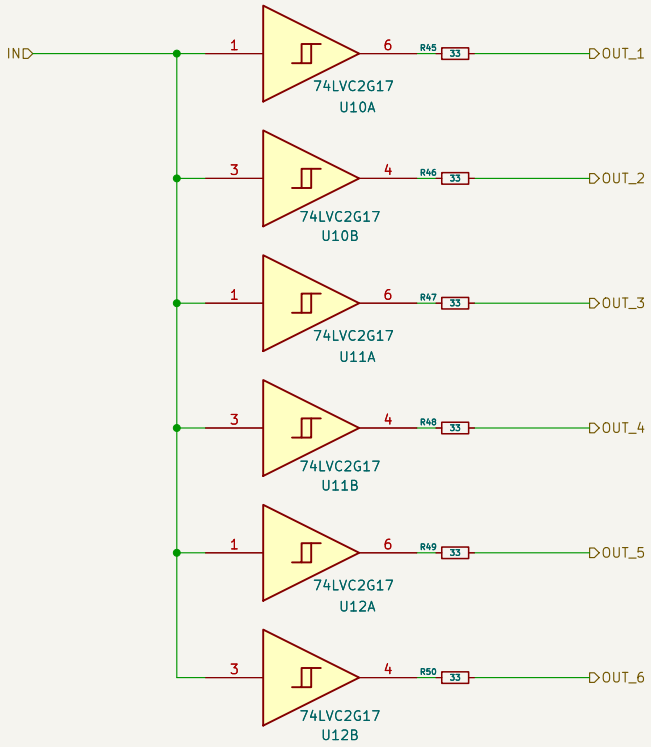
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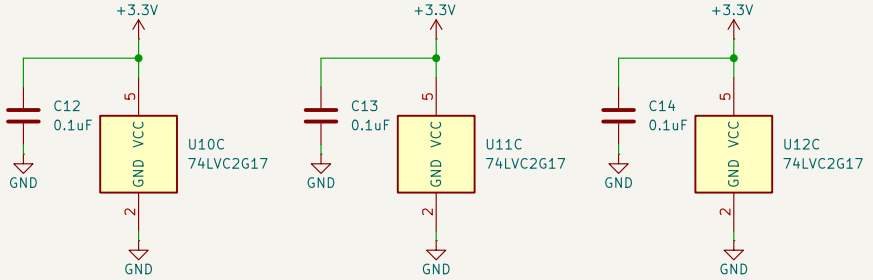
Sheet: /Fan Out/COPI B0 fan out to P1–P6/

File: fan_out_by_5x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 5/67



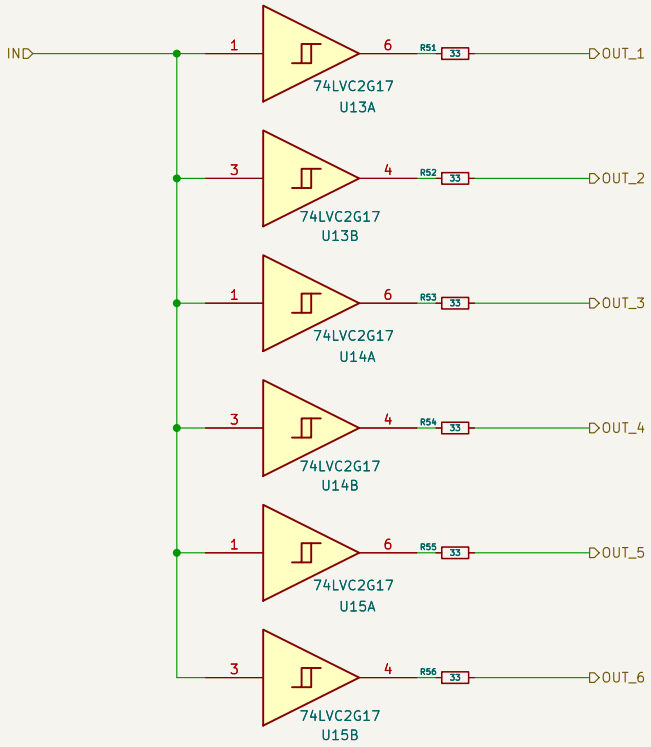


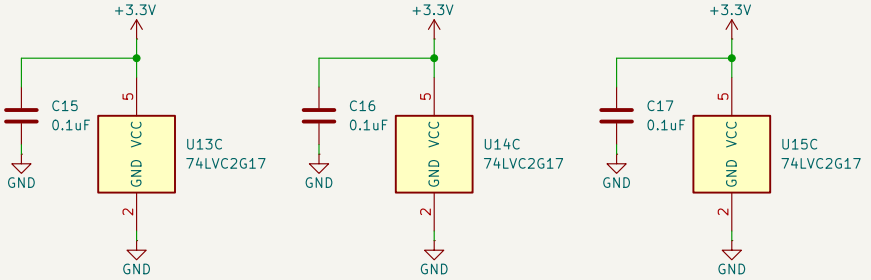
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Sheet: /Fan Out/SCK B1 fan out to P6–P10/
File: fan_out_by_5x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 6/67



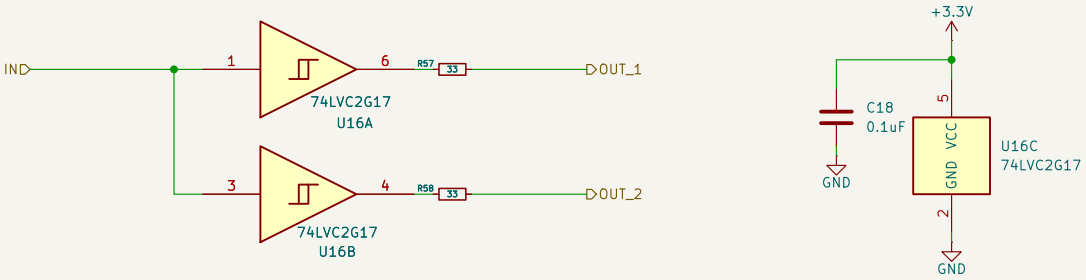


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Sheet: /Fan Out/COPI B1 fan out to P6-P10/
File: fan_out_by_5x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 7/67



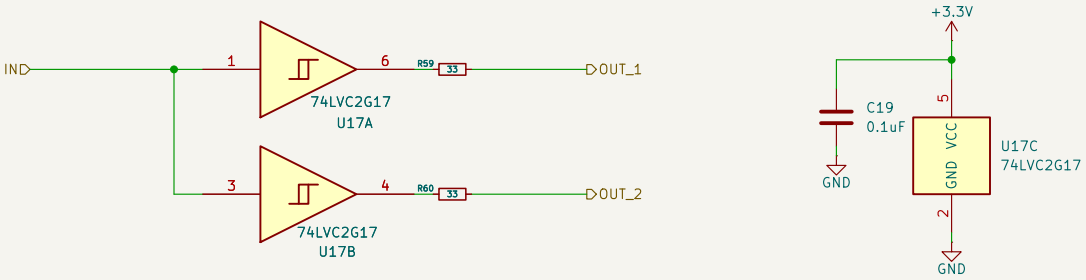
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Sheet: /Fan Out/CS_00 fan out to P1, P6/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 8/67



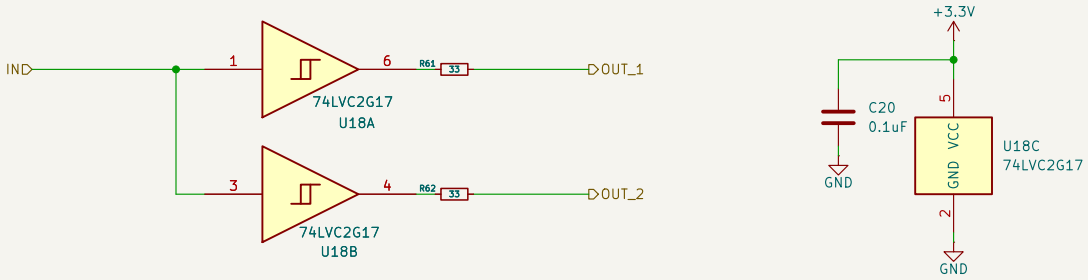
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Sheet: /Fan Out/CS_01 fan out to P1, P6/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 9/67



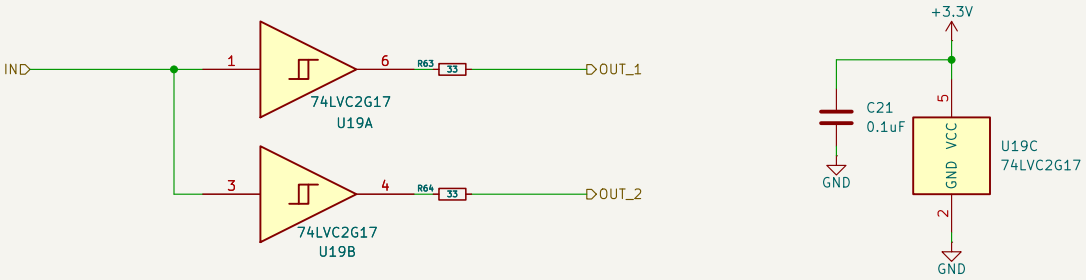
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Sheet: /Fan Out/CS_02 fan out to P1, P6/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 10/67



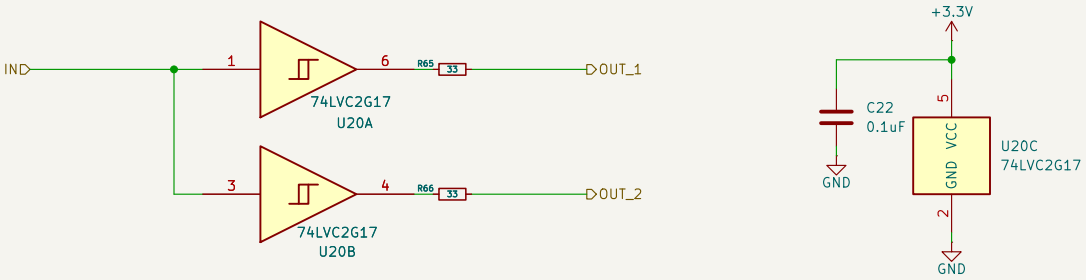
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Sheet: /Fan Out/CS_03 fan out to P1, P6/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 11/67



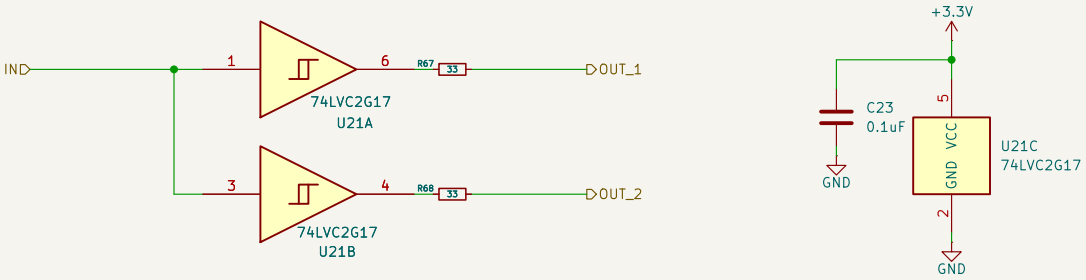
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Sheet: /Fan Out/CS_04 fan out to P2, P7/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 12/67



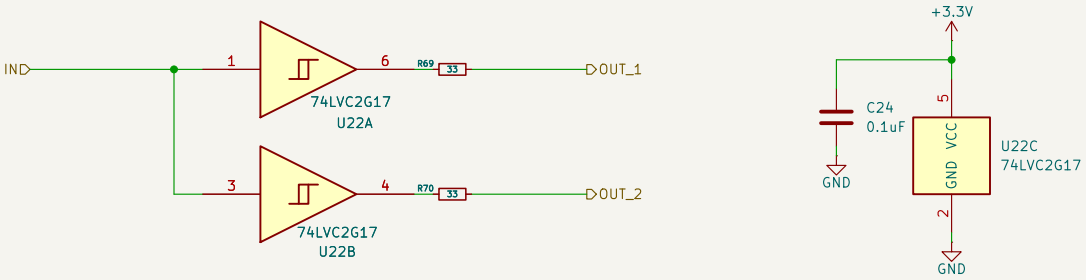
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Sheet: /Fan Out/CS_05 fan out to P2, P7/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 13/67

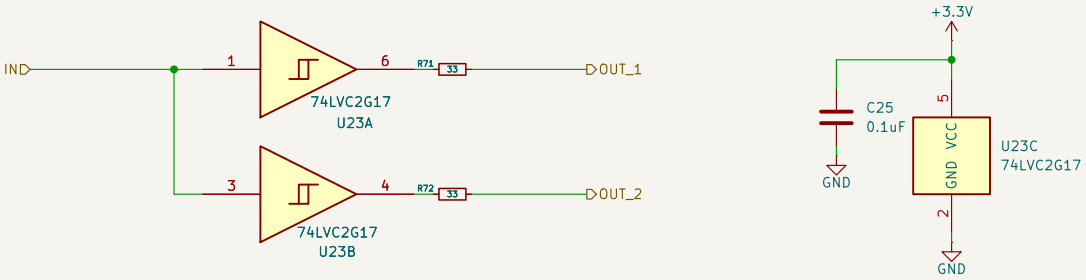


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Sheet: /Fan Out/CS_06 fan out to P2, P7/
File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 14/67



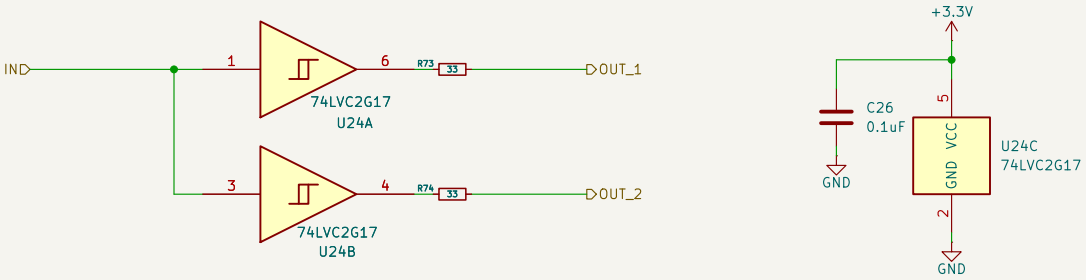
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Sheet: /Fan Out/CS_07 fan out to P2, P7/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 15/67

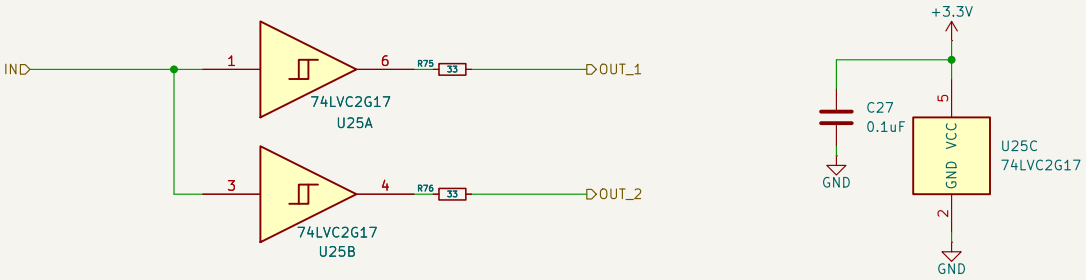


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Sheet: /Fan Out/CS_08 fan out to P3, P8/
File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 16/67

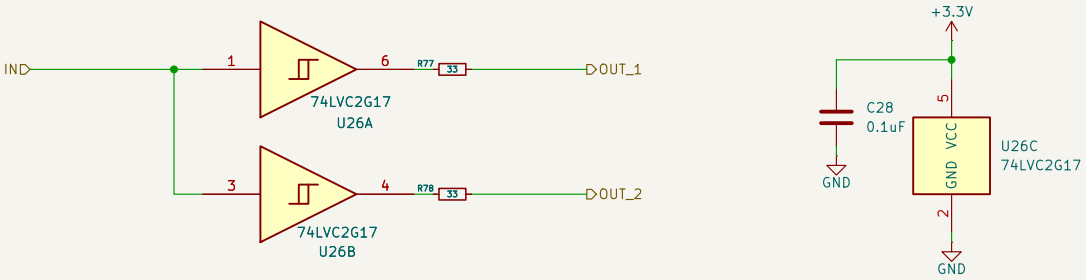


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Sheet: /Fan Out/CS_09 fan out to P3, P8/
File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 17/67

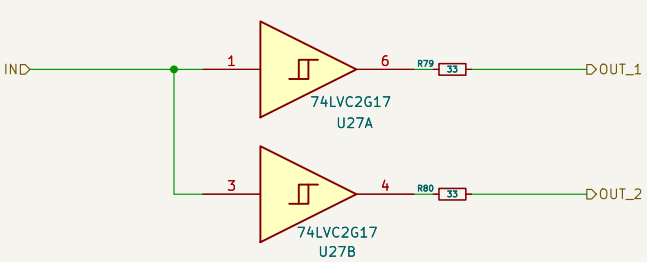


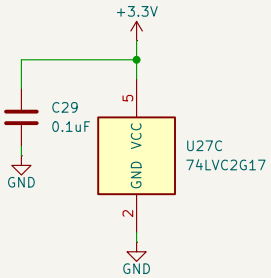
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Sheet: /Fan Out/CS_10 fan out to P3, P8/
File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
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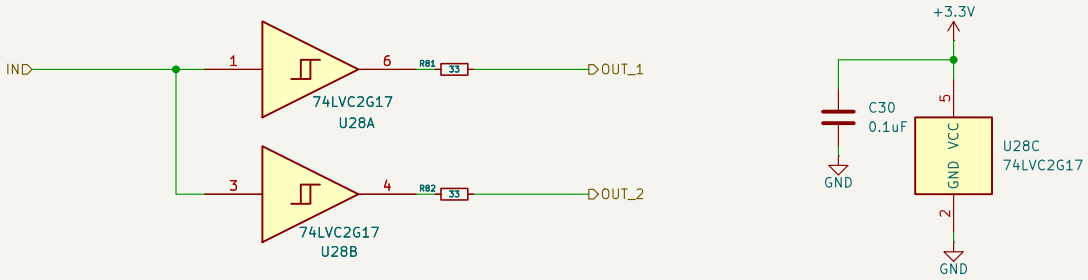
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Sheet: /Fan Out/CS_11 fan out to P3, P8/

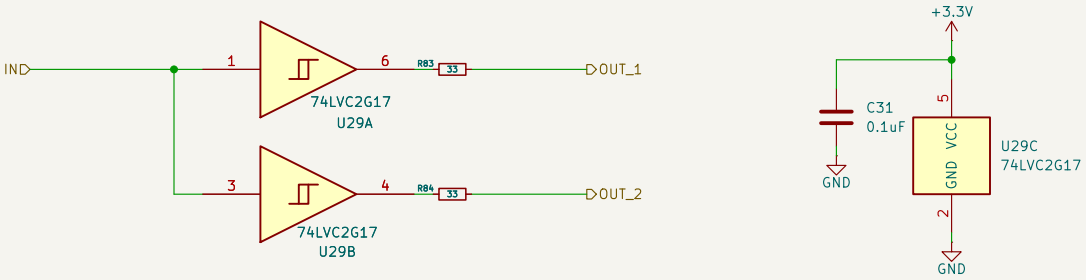
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Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 19/67



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Sheet: /Fan Out/CS_12 fan out to P4, P9/	
File: fan_out_by_2x.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 20/67

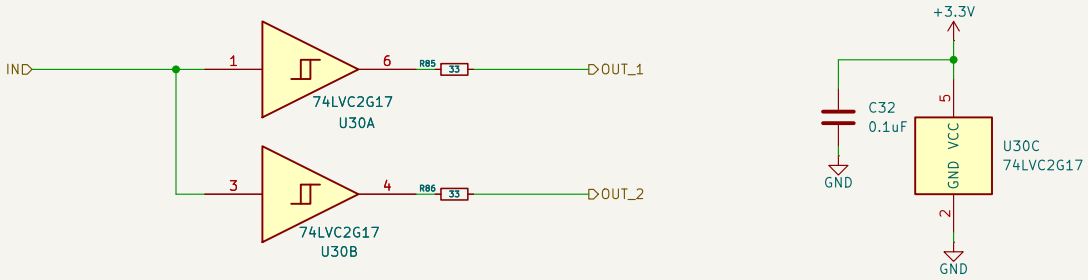


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Sheet: /Fan Out/CS_13 fan out to P4, P9/
File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 21/67



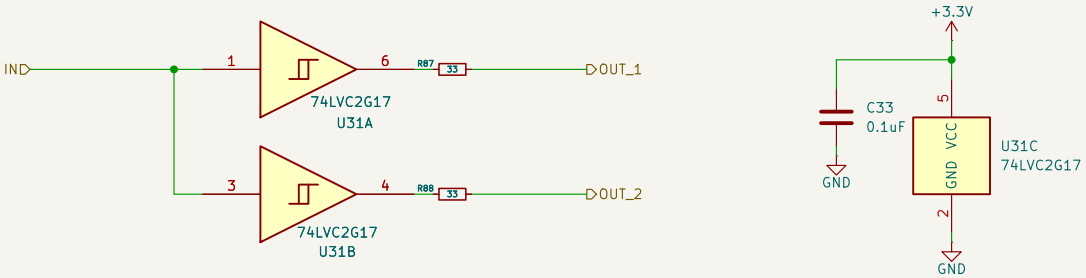
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Sheet: /Fan Out/CS_14 fan out to P4, P9/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 22/67



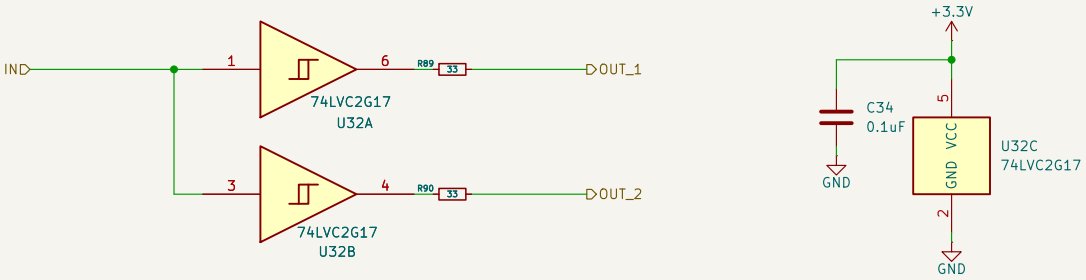
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Sheet: /Fan Out/CS_15 fan out to P4, P9/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 23/67

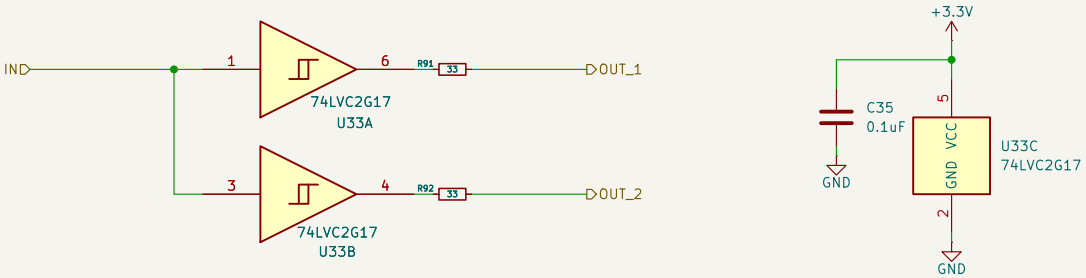


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Sheet: /Fan Out/CS_16 fan out to P5, P10/
File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 24/67

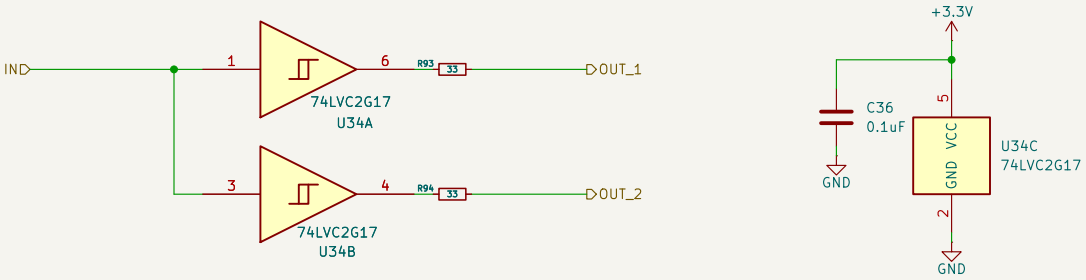


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Sheet: /Fan Out/CS_17 fan out to P5, P10/
File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 25/67



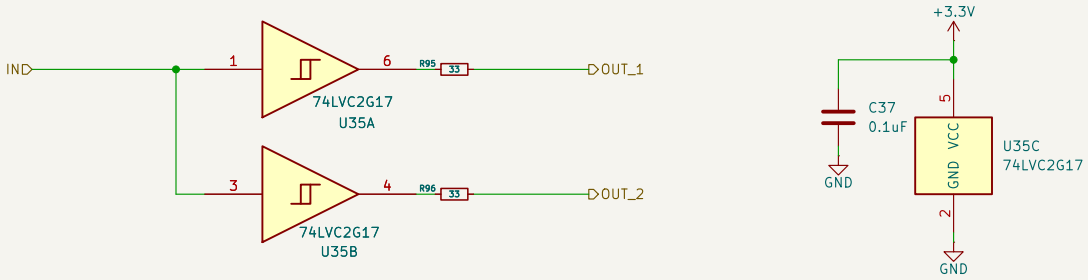
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Sheet: /Fan Out/CS_18 fan out to P5, P10/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 26/67



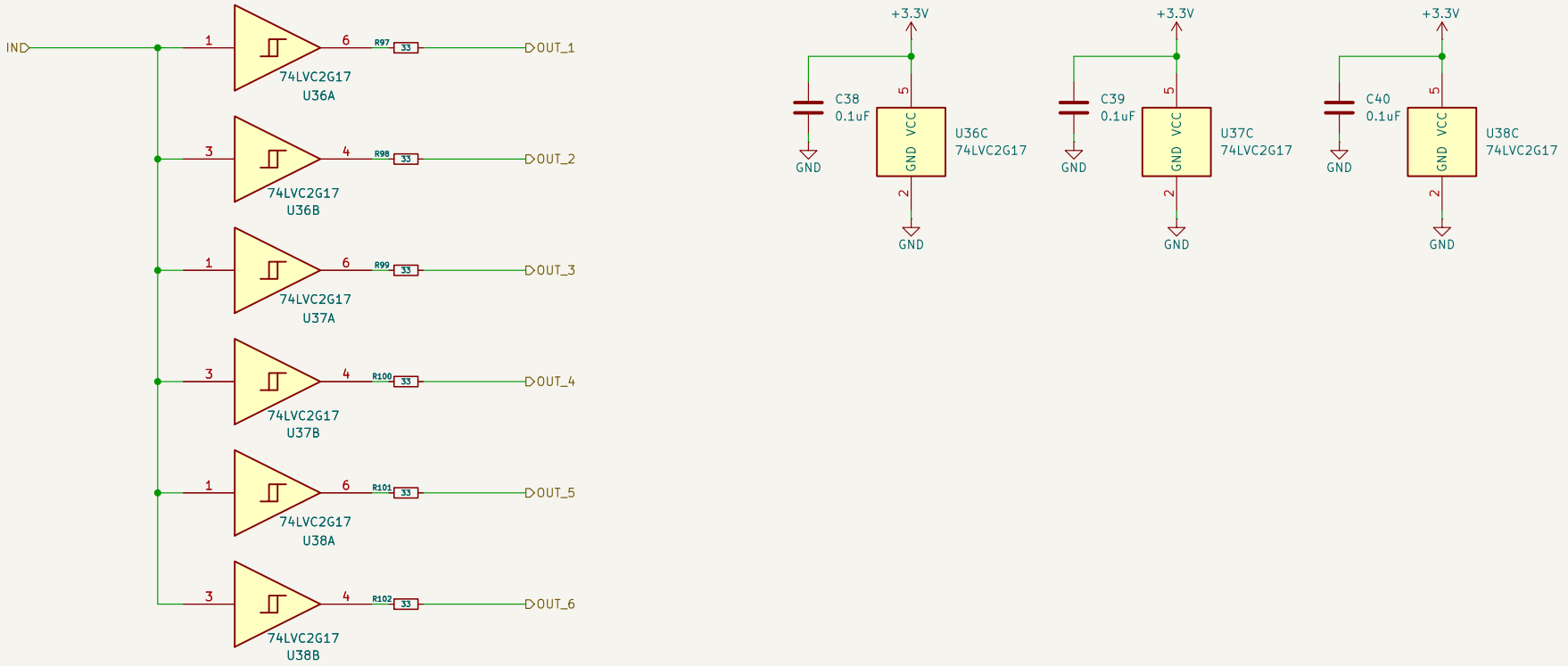
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Sheet: /Fan Out/CS_19 fan out to P5, P10/

File: fan_out_by_2x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 27/67

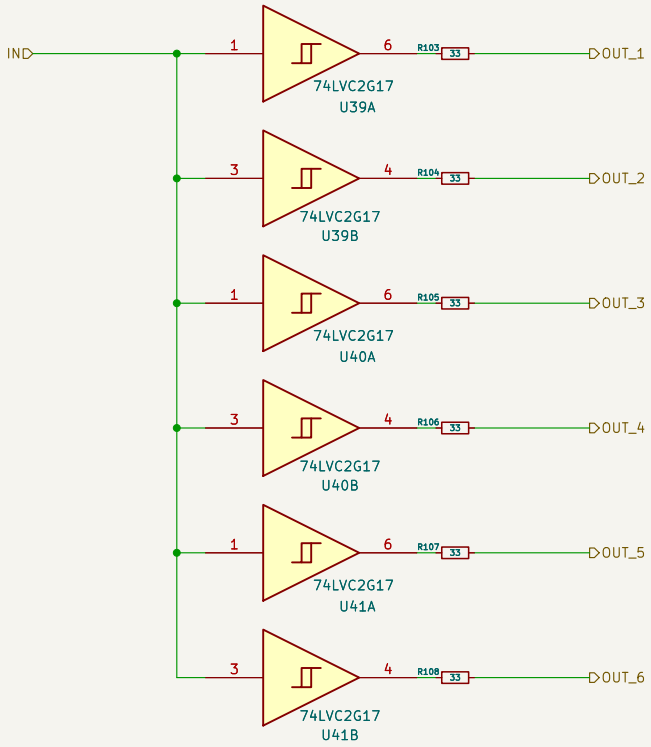


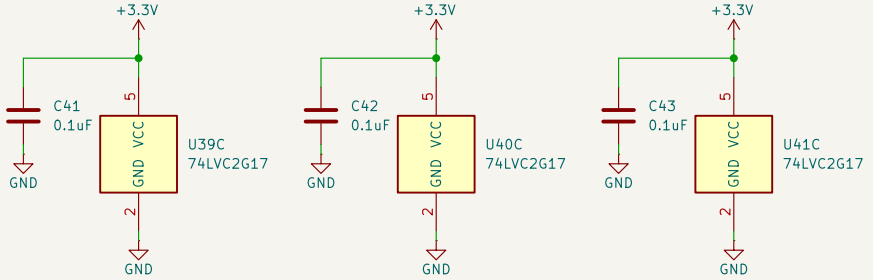
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Sheet: /Fan Out/EINT fan out to P6–P10/
File: fan_out_by_5x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 28/67



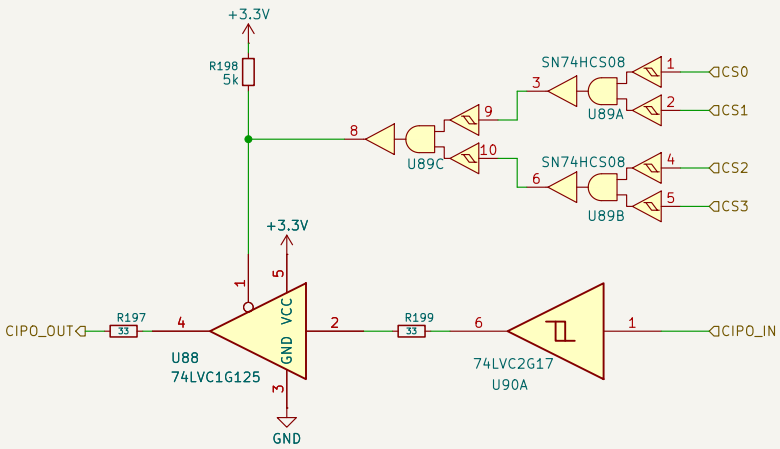


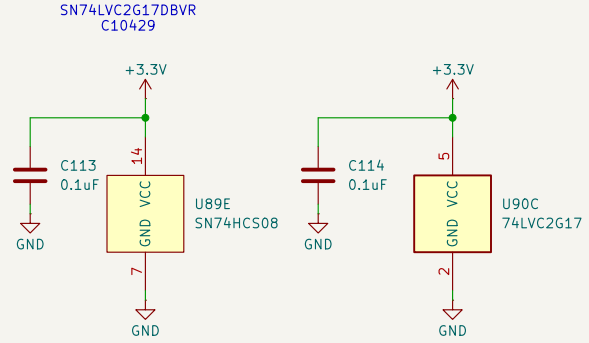
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Sheet: /Fan Out/EINT fan out to P1–P5/
File: fan_out_by_5x.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 29/67







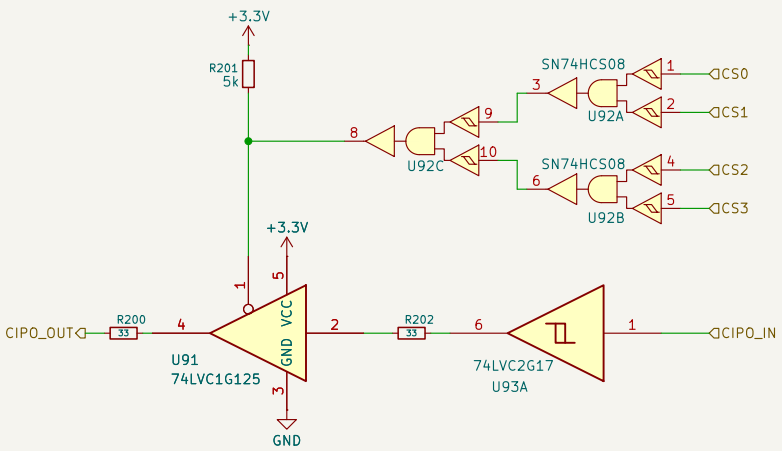
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Sheet: /Fan Out/CIPO B0 Enable for P1/

File: cipo_enable.kicad_sch

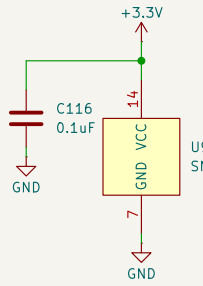
Title: G6 Teensy 10-14 Arena

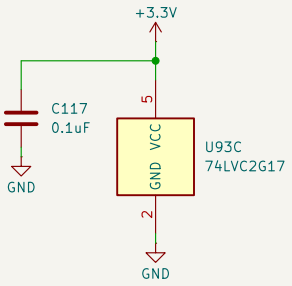
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 54/67

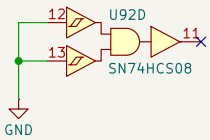


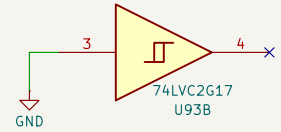


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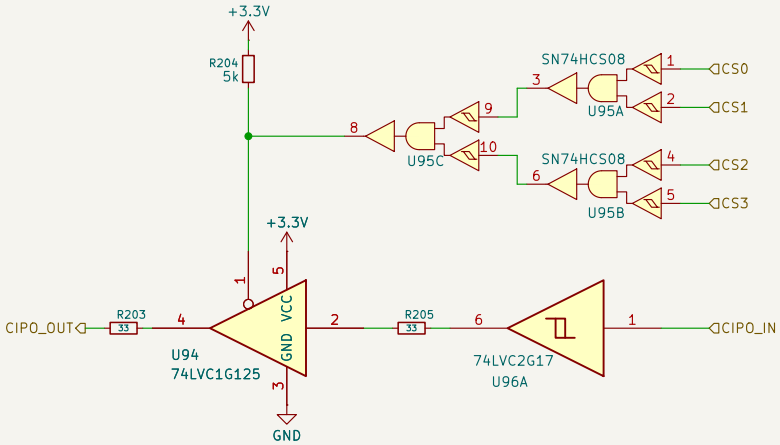


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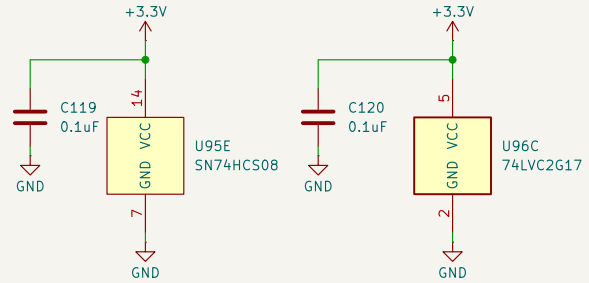
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File: cipo_enable.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 55/67



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C10429



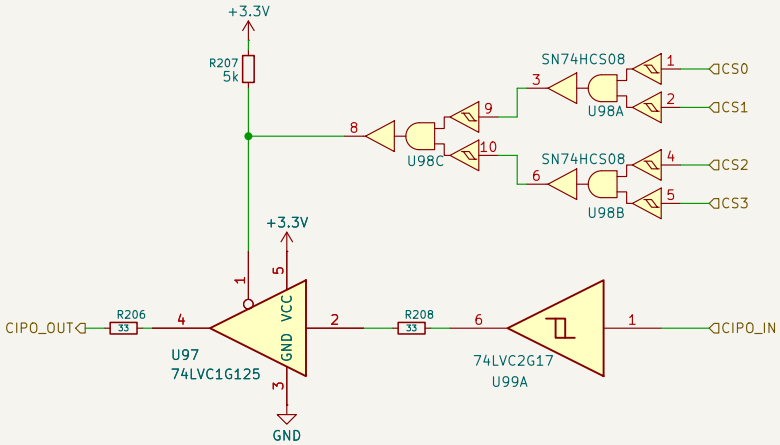


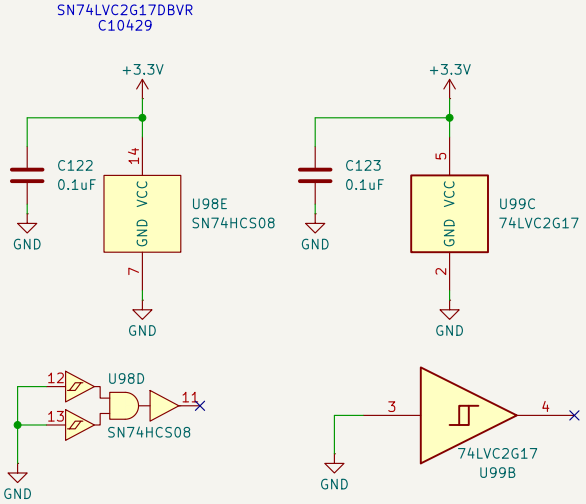
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Sheet: /Fan Out/CIPO B0 Enable for P3/
File: cipo_enable.kicad_sch

Title: G6 Teensy 10-14 Arena

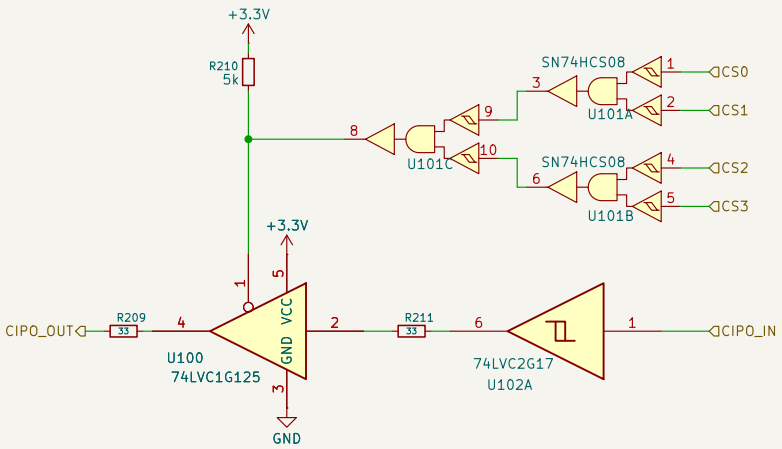
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 56/67



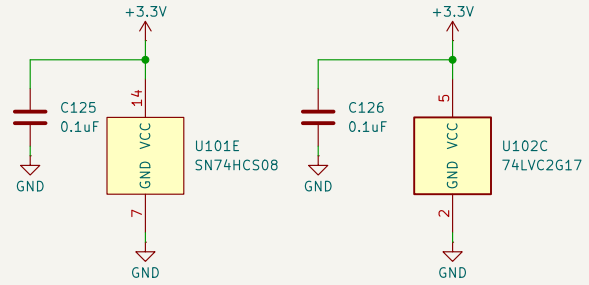


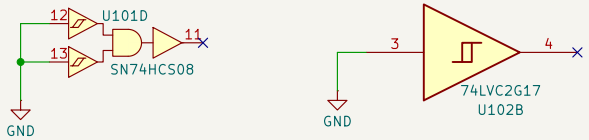


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Sheet: /Fan Out/CIPO B0 Enable for P4/	
File: cipo_enable.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 57/67



SN74LVC2G17DBVR
C10429





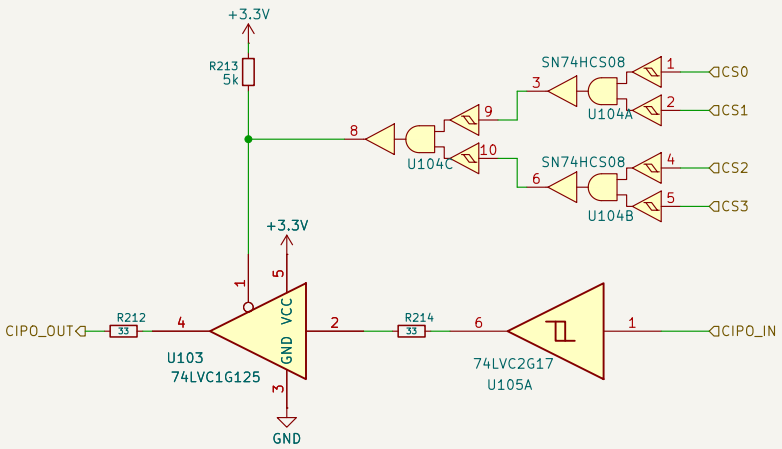


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Sheet: /Fan Out/CIPO B0 Enable for P5/
File: cipo_enable.kicad_sch

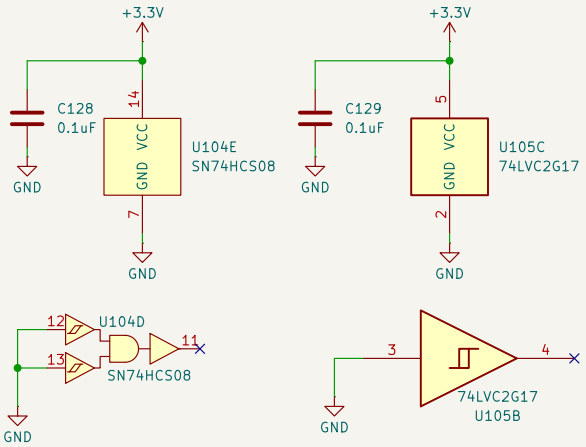
Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 58/67

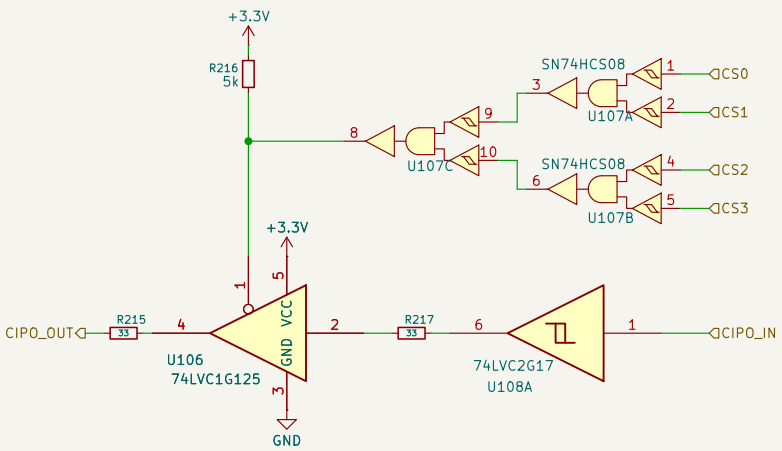


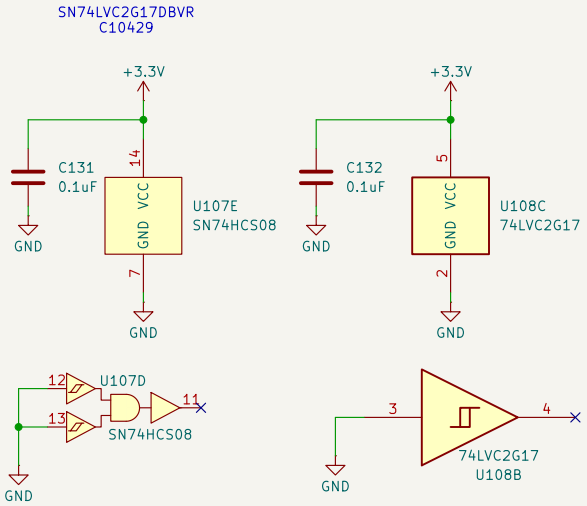


SN74LVC2G17DBVR
C10429



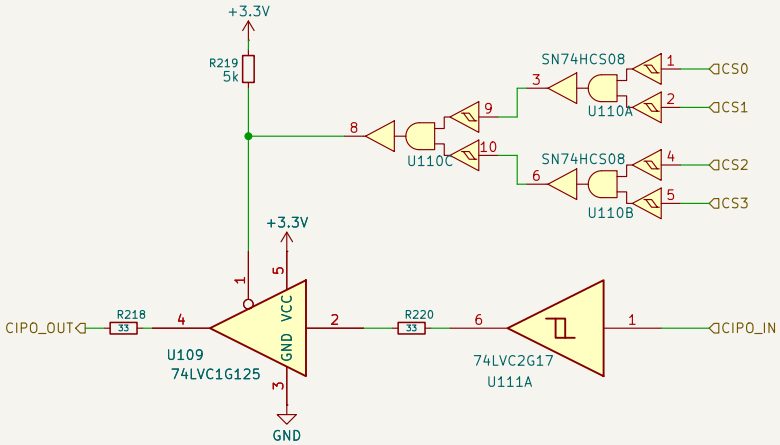
Reiser Lab @ Janelia | HHMI	
Sheet: /Fan Out/CIPO B1 Enable for P7/	
File: cipo_enable.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 59/67





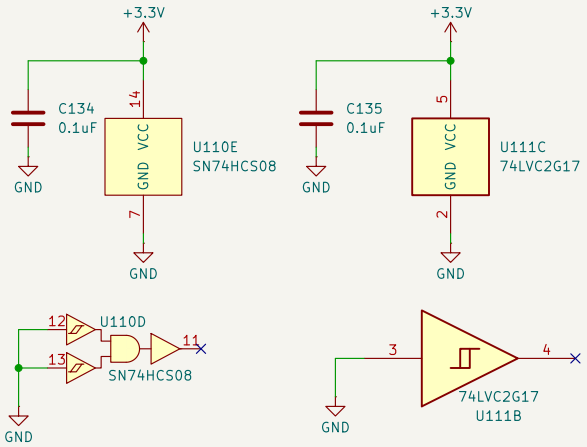


Reiser Lab @ Janelia | HHMI	
Sheet: /Fan Out/CIPO B1 Enable for P8/	
File: cipo_enable.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 60/67





SN74LVC2G17DBVR
C10429



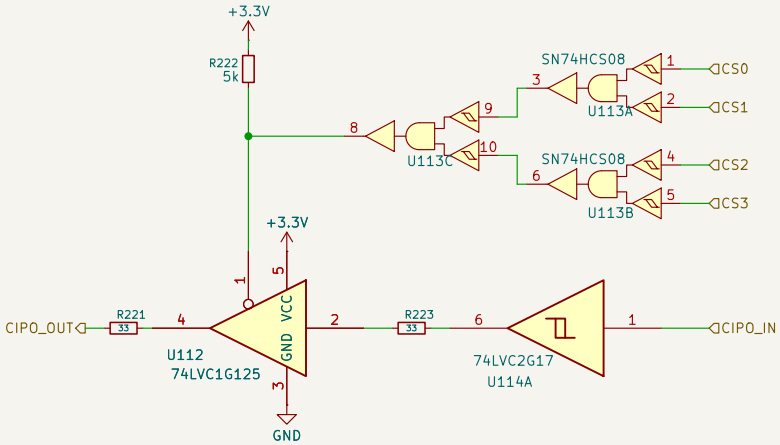
Reiser Lab @ Janelia | HHMI

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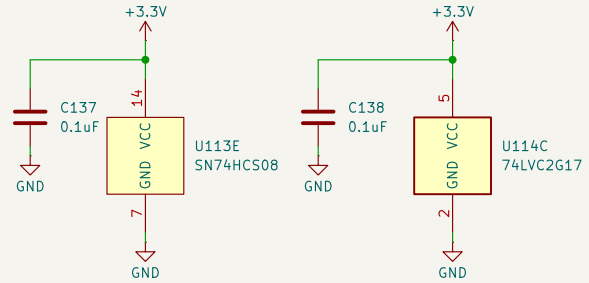
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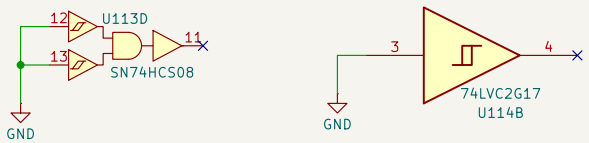
Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 61/67



SN74LVC2G17DBVR
C10429





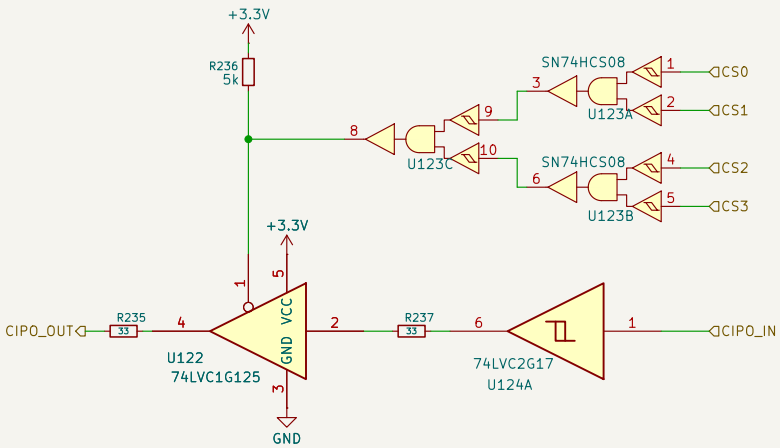


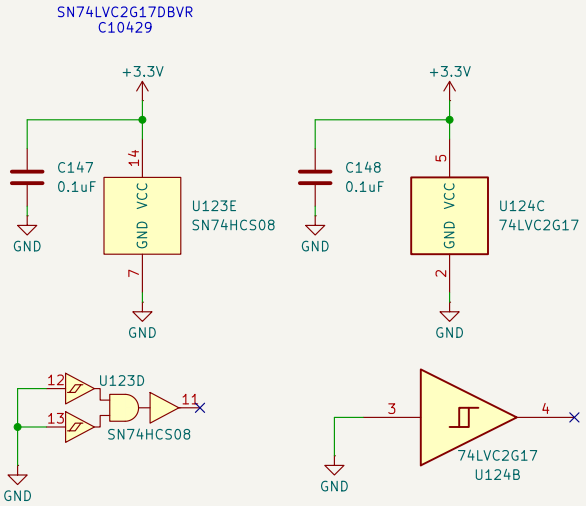
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Sheet: /Fan Out/CIPO B1 Enable for P10/
File: cipo_enable.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 62/67







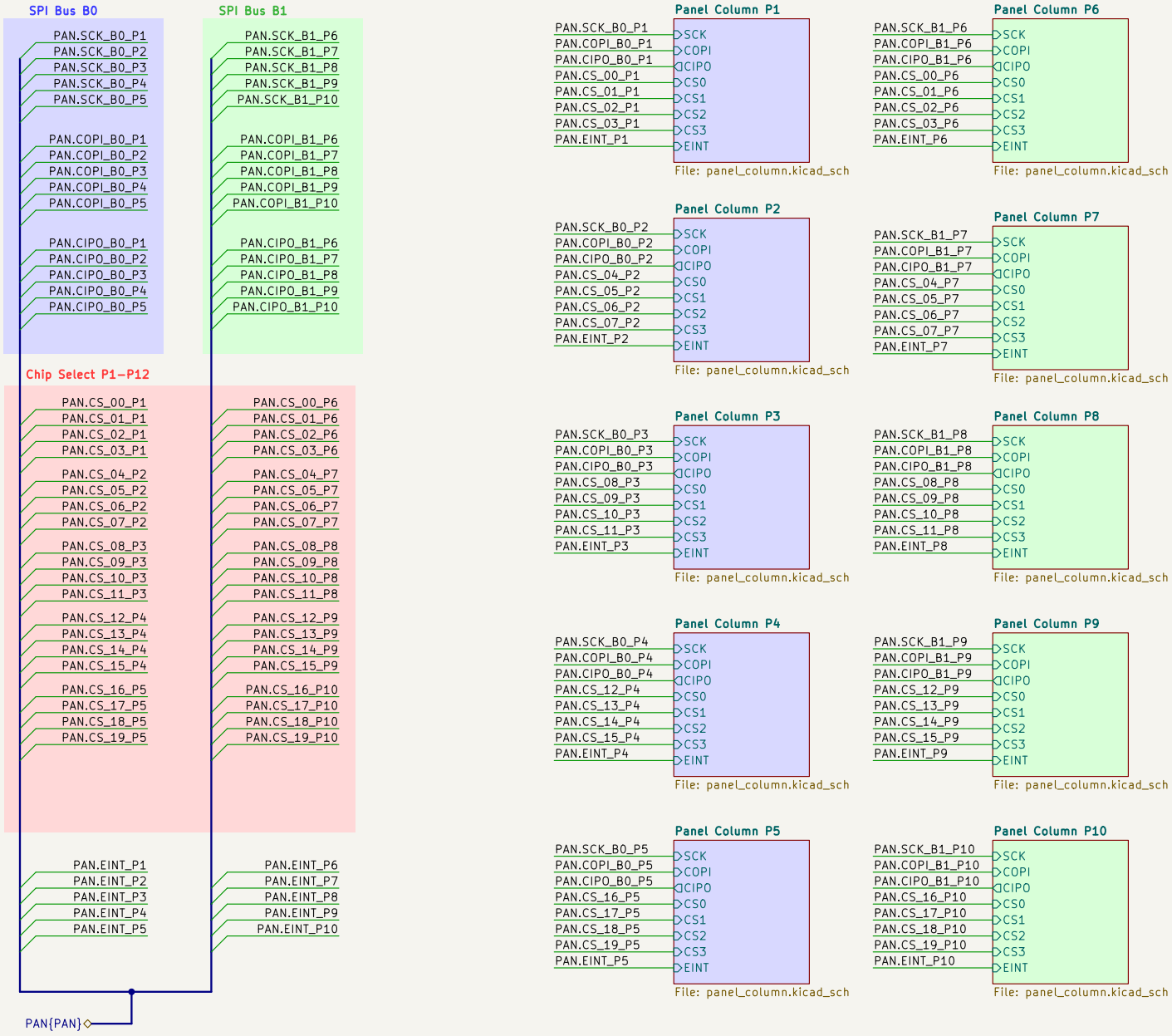
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Sheet: /Fan Out/CIPO B1 Enable for P6/

File: cipo_enable.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 68/67



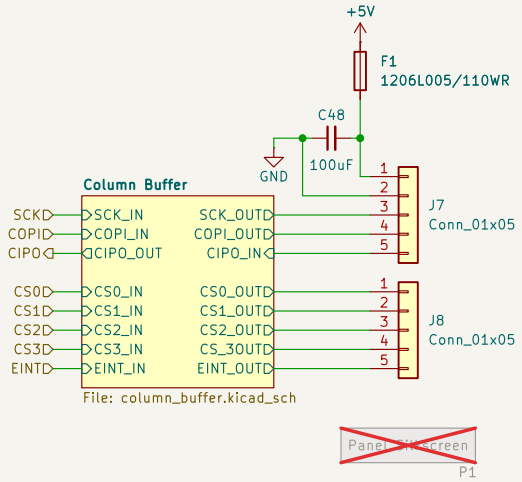
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Sheet: /Panels/

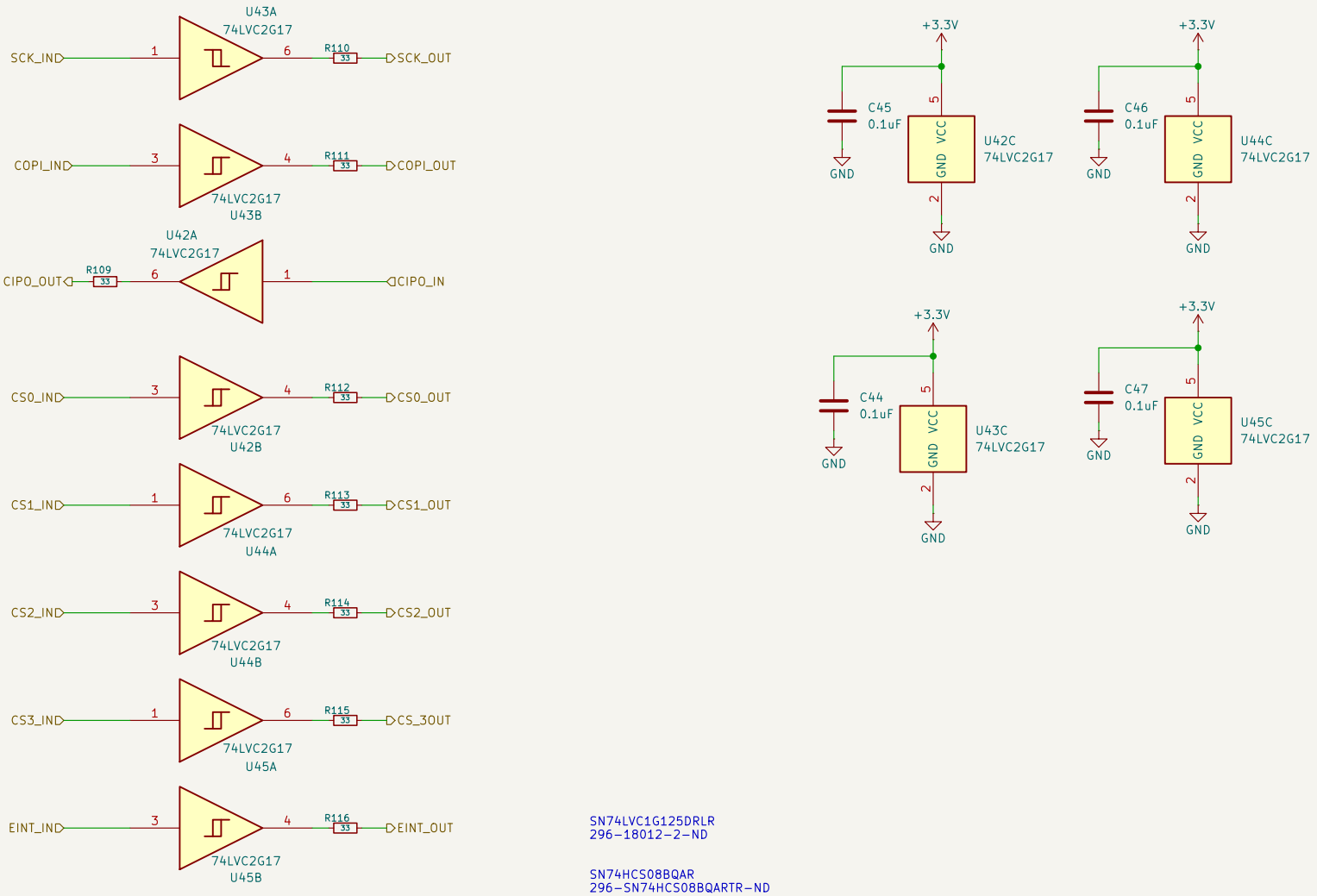
File: panels.kicad_sch

Title: G6 Teensy 10–14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 30/67



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Sheet: /Panels/Panel Column P1/	
File: panel_column.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 32/67



SN74LVC1G125DRLR
296-18012-2-ND

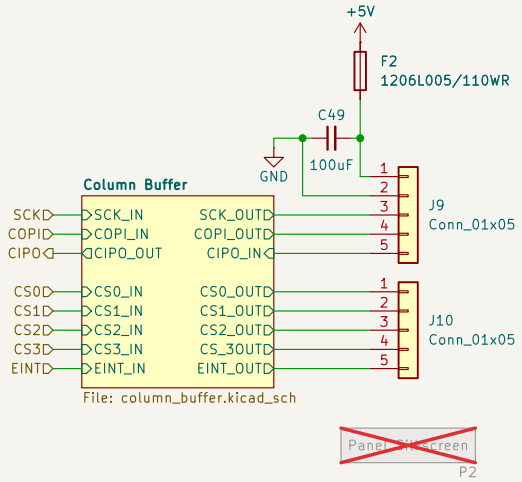
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

Reiser Lab @ Janelia | HHMI

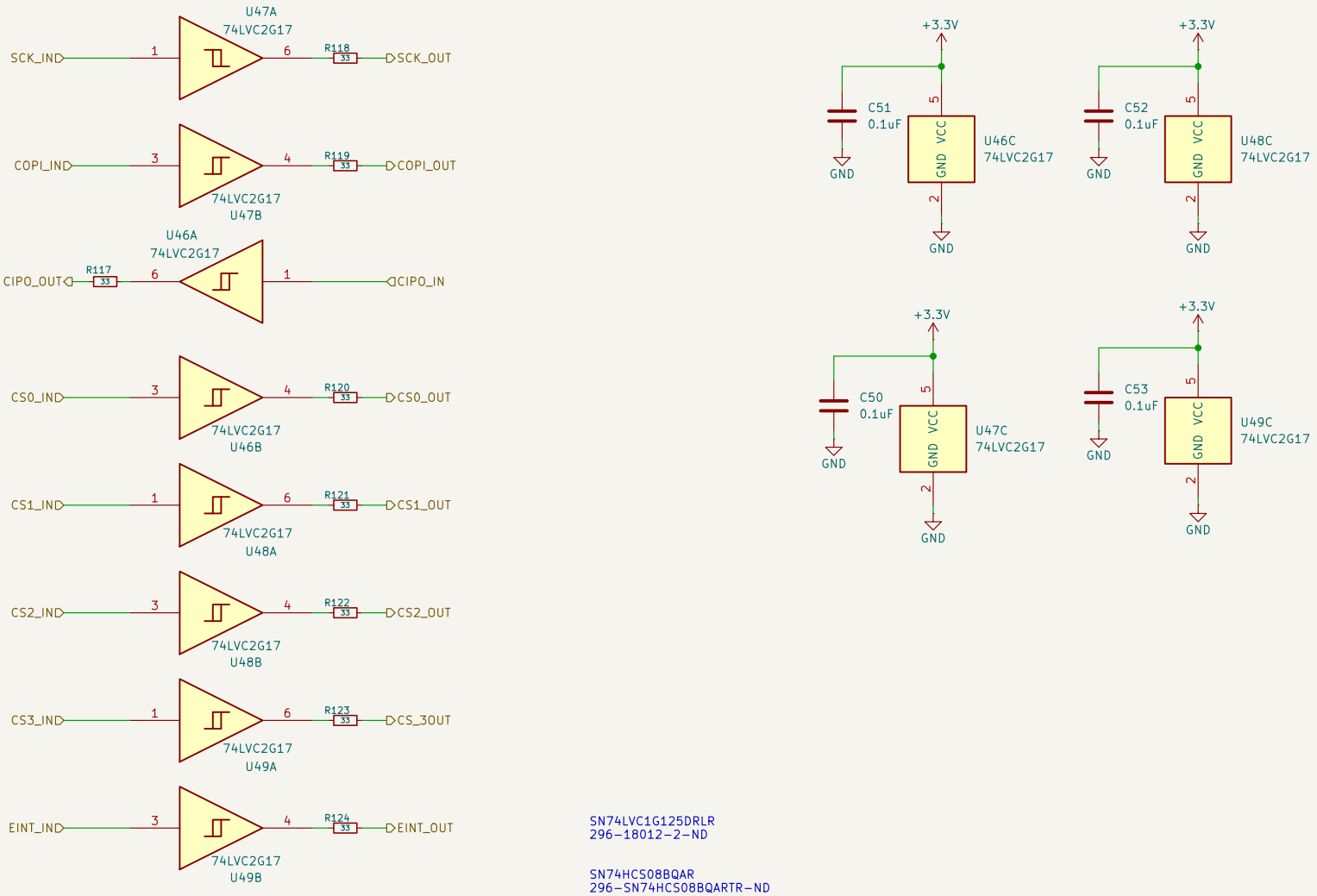
Sheet: /Panels/Panel Column P1/Column Buffer/
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 31/67



Reiser Lab @ Janelia | HHMI	
Sheet: /Panels/Panel Column P2/	
File: panel_column.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 33/67



SN74LVC1G125DRLR
296-18012-2-ND

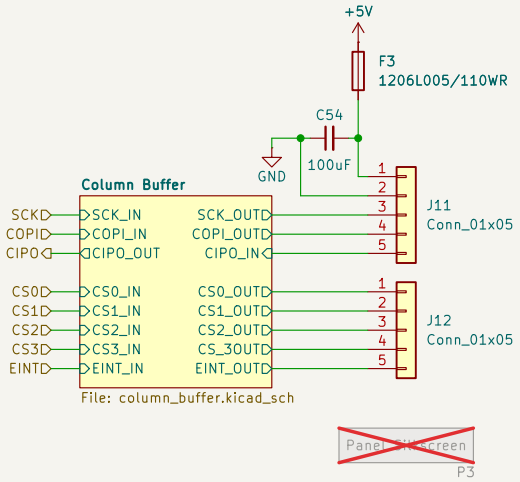
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

Reiser Lab @ Janelia | HHMI

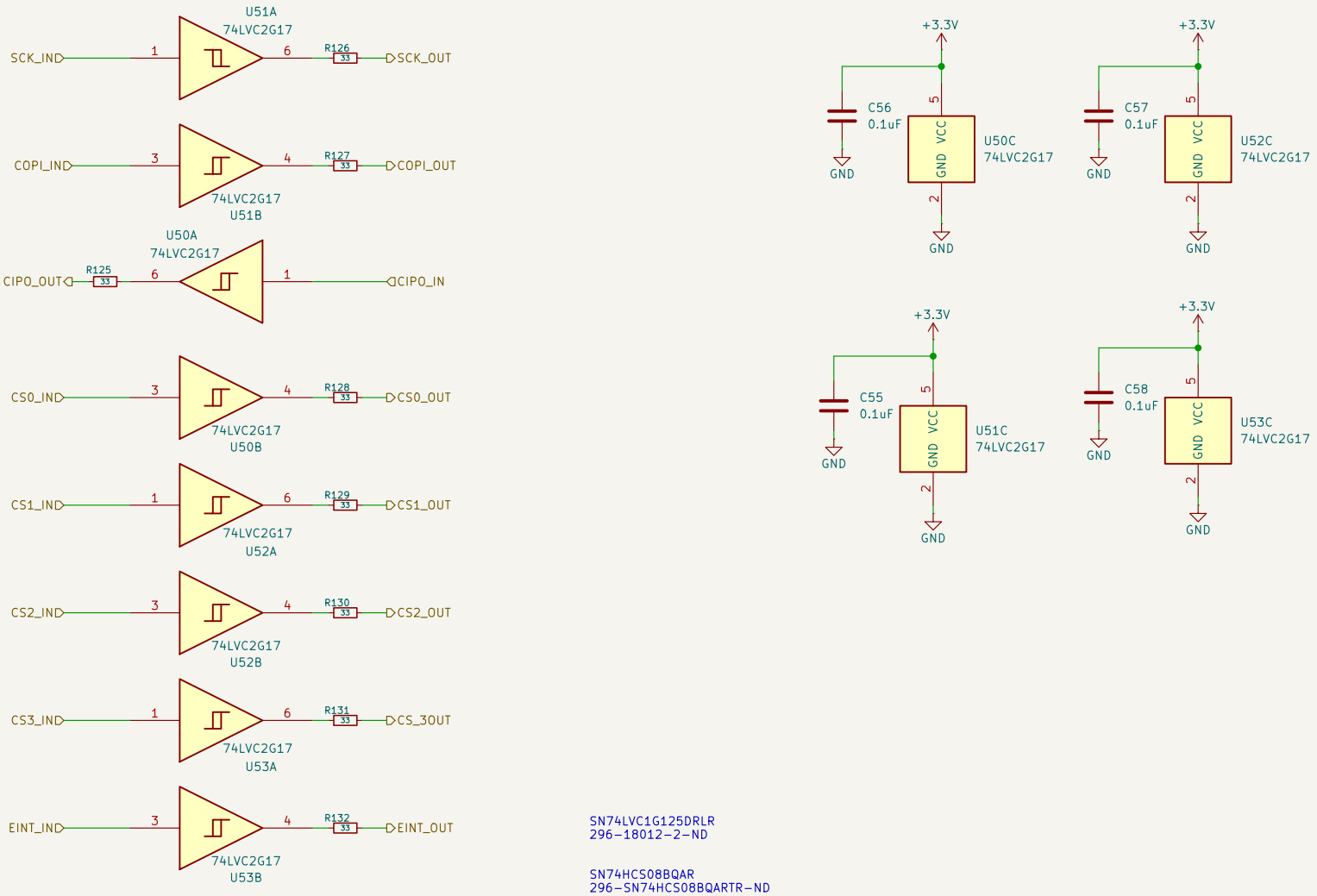
Sheet: /Panels/Panel Column P2/Column Buffer/
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 34/67



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Sheet: /Panels/Panel Column P3/		
File: panel_column.kicad_sch		
Title: G6 Teensy 10–14 Arena		
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 35/67



SN74LVC1G125DRLR
296-18012-2-ND

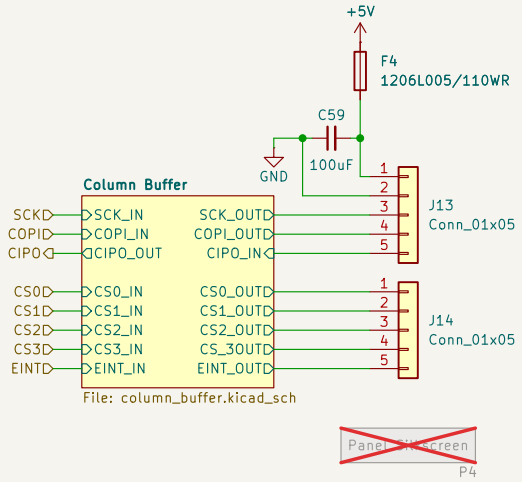
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

Reiser Lab @ Janelia | HHMI

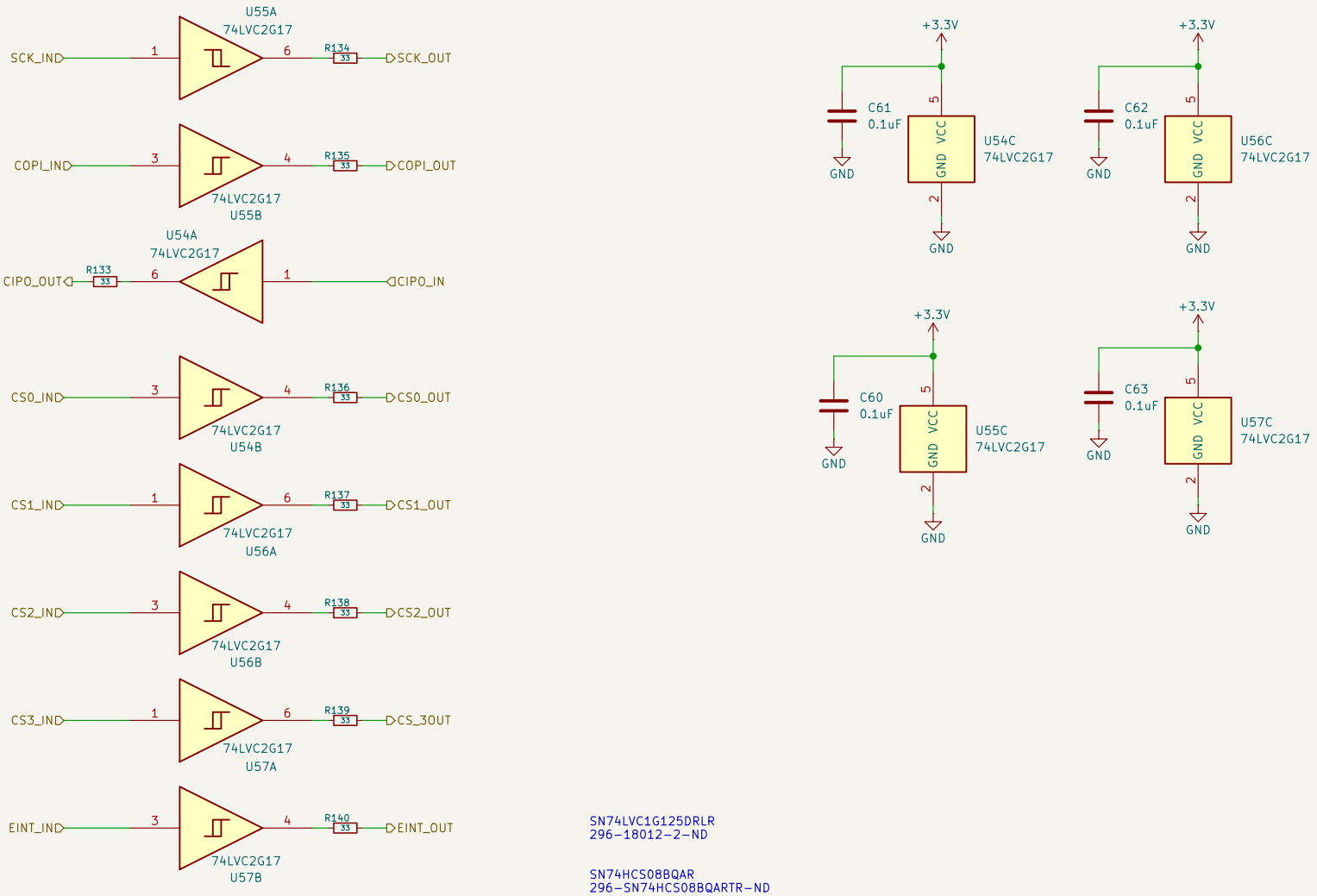
Sheet: /Panels/Panel Column P3/Column Buffer/
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 36/67



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Sheet: /Panels/Panel Column P4/		
File: panel_column.kicad_sch		
Title: G6 Teensy 10–14 Arena		
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 37/67



SN74LVC1G125DRLR
296-18012-2-ND

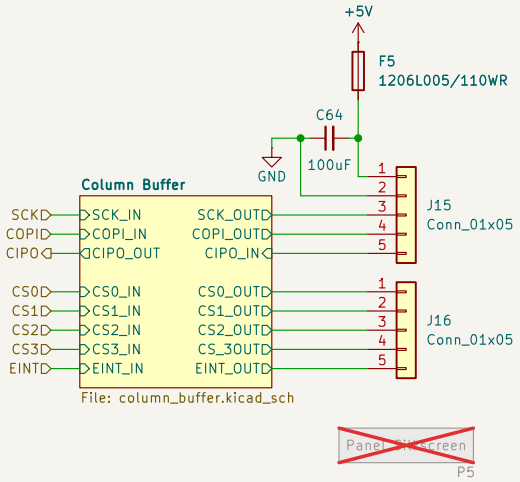
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

Reiser Lab @ Janelia | HHMI

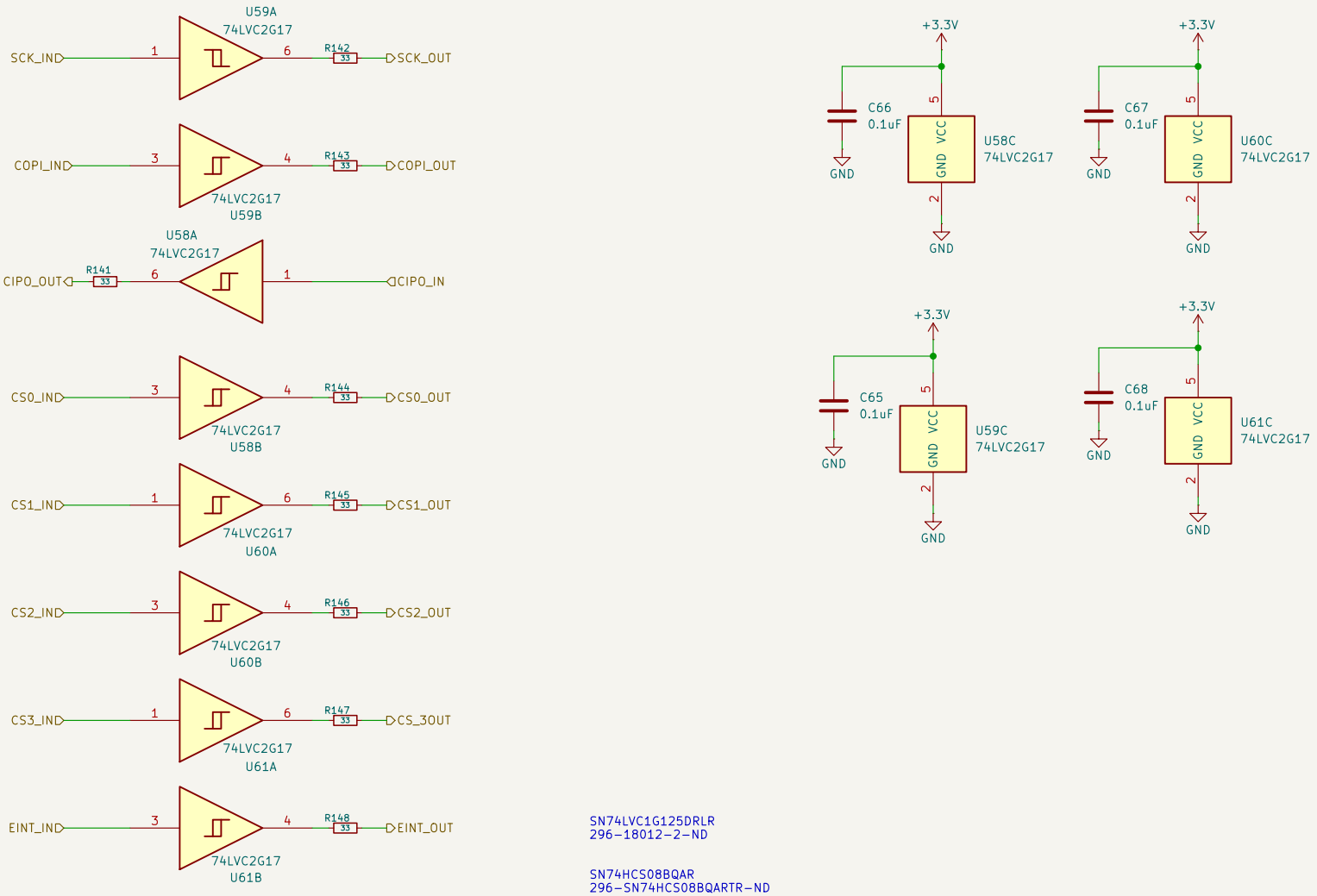
Sheet: /Panels/Panel Column P4/Column Buffer/
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 38/67



Reiser Lab @ Janelia | HHMI	
Sheet: /Panels/Panel Column P5/	
File: panel_column.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 39/67



SN74LVC1G125DRLR
296-18012-2-ND

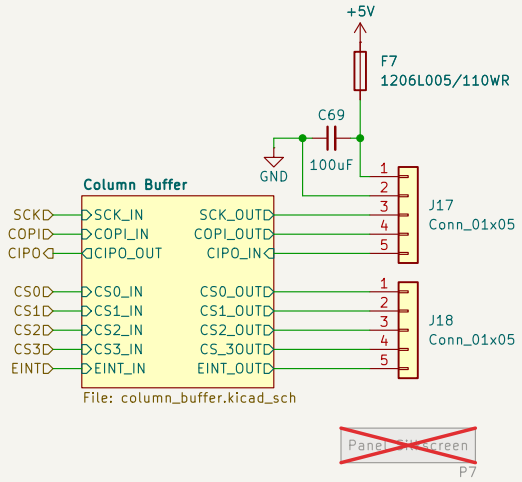
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

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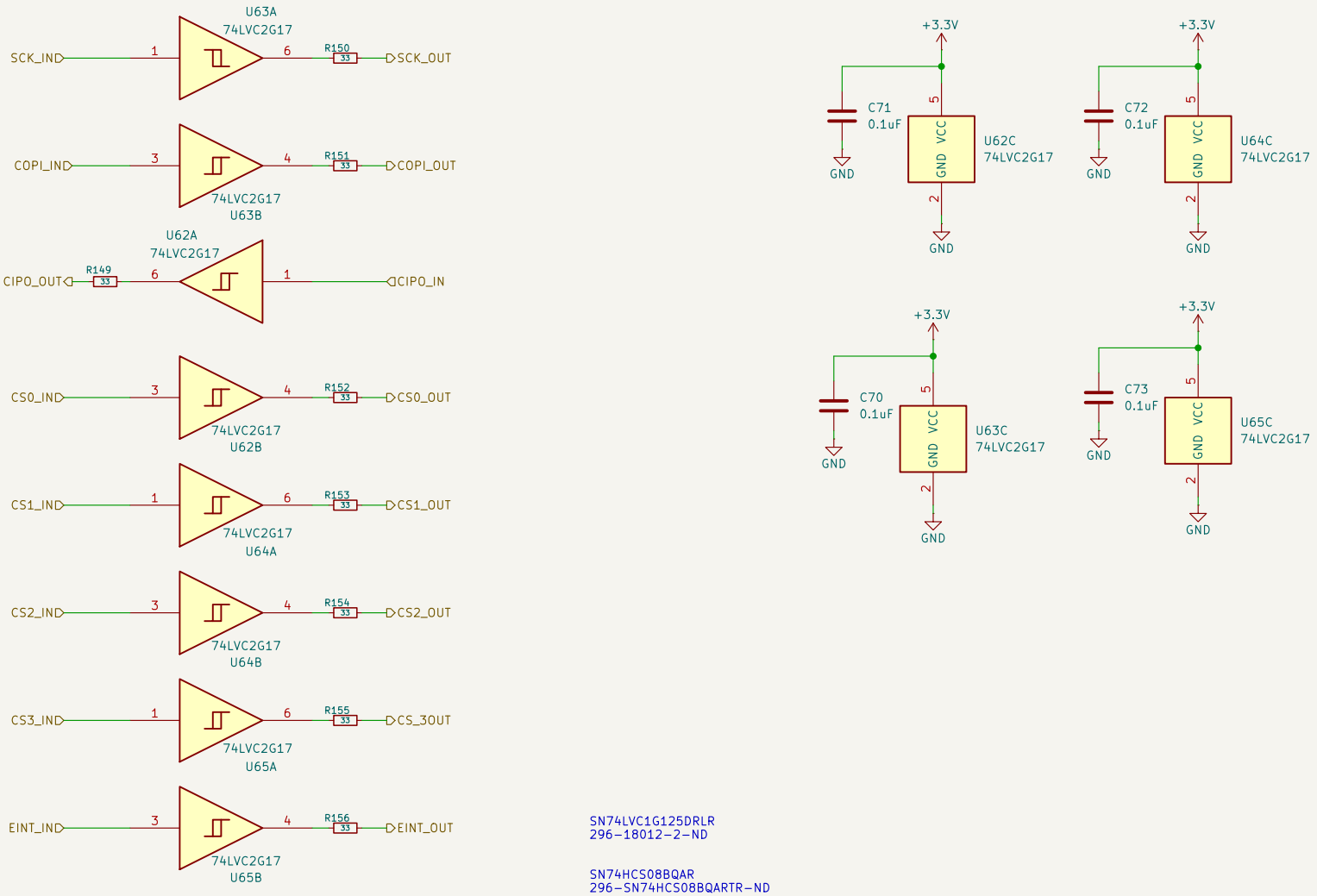
Sheet: /Panels/Panel Column P5/Column Buffer/
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 40/67



Reiser Lab @ Janelia | HHMI	
Sheet: /Panels/Panel Column P7/	
File: panel_column.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 41/67



SN74LVC1G125DRLR
296-18012-2-ND

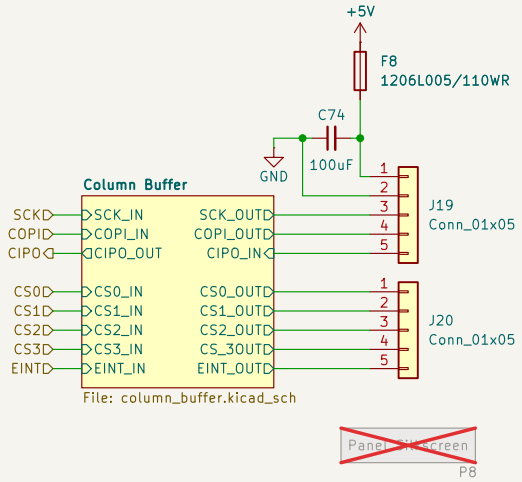
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

Reiser Lab @ Janelia | HHMI

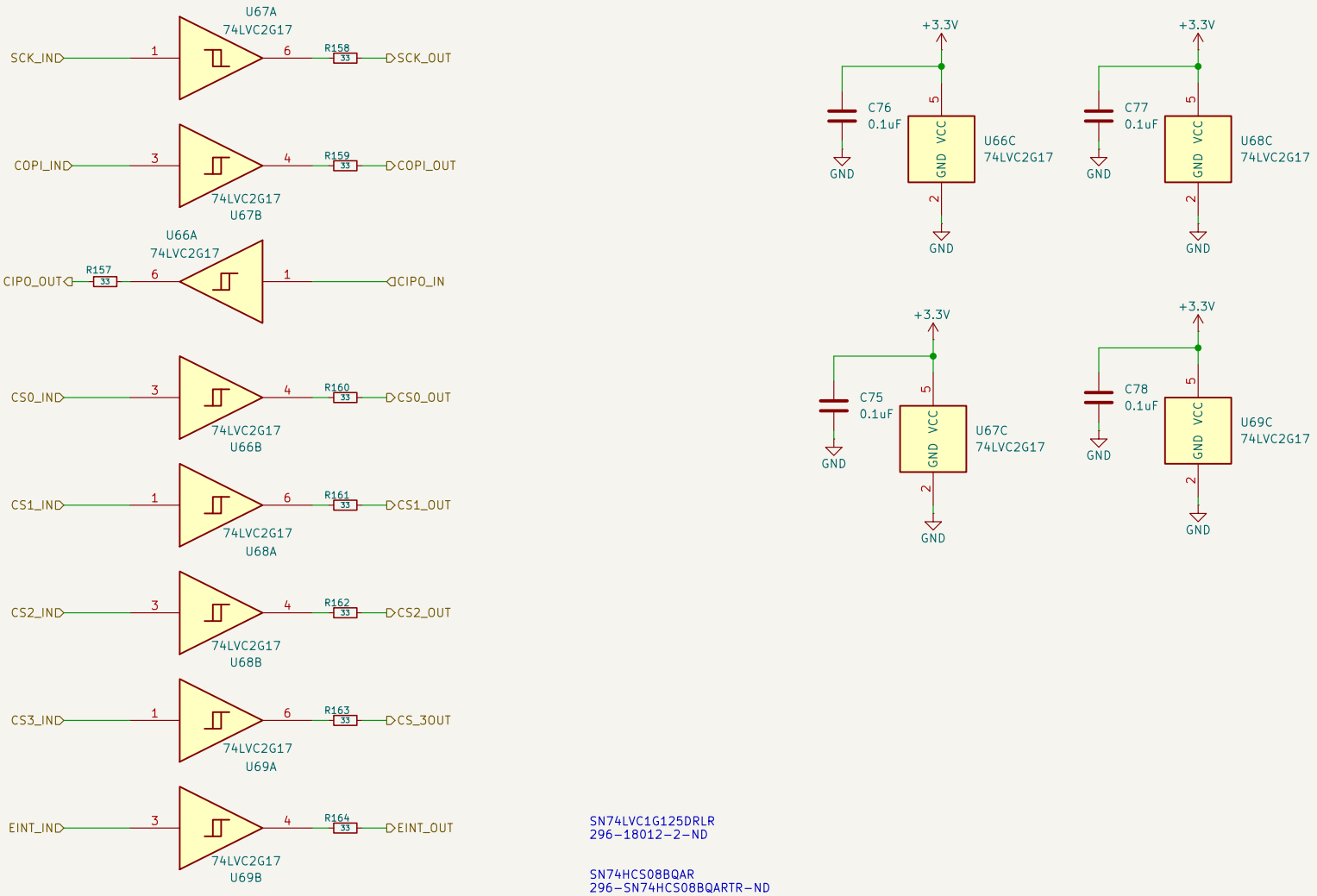
Sheet: /Panels/Panel Column P7/Column Buffer/
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 42/67



Reiser Lab @ Janelia | HHMI	
Sheet: /Panels/Panel Column P8/	
File: panel_column.kicad_sch	
Title: G6 Teensy 10–14 Arena	
Size: USLetter | Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7	Id: 43/67



SN74LVC1G125DRLR
296-18012-2-ND

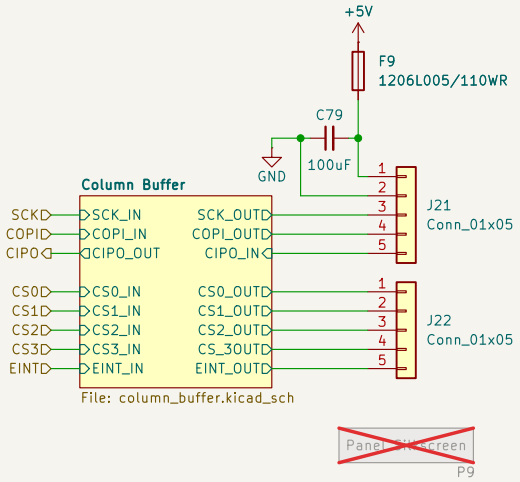
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

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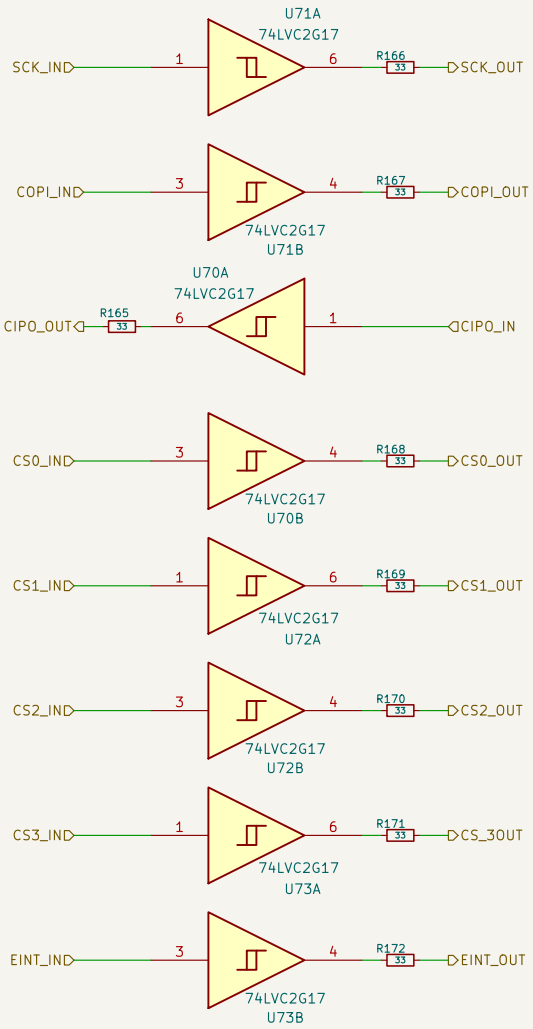
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File: column_buffer.kicad_sch

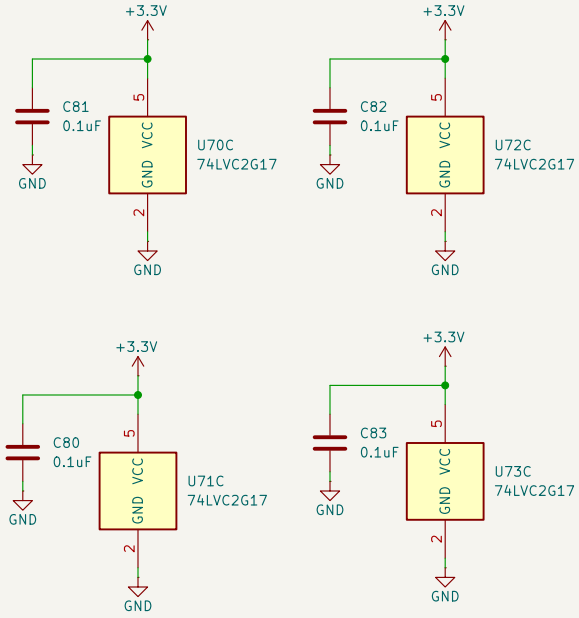
Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 44/67



Reiser Lab @ Janelia | HHMI		
Sheet: /Panels/Panel Column P9/		
File: panel_column.kicad_sch		
Title: G6 Teensy 10–14 Arena		
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 45/67





SN74LVC1G125DRLR
296-18012-2-ND

SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

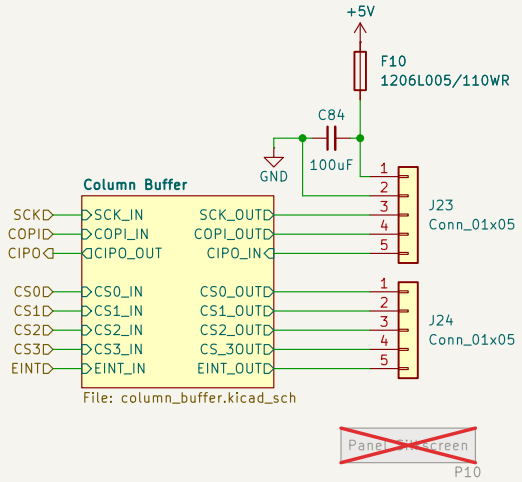
Reiser Lab @ Janelia | HHMI

Sheet: /Panels/Panel Column P9/Column Buffer/

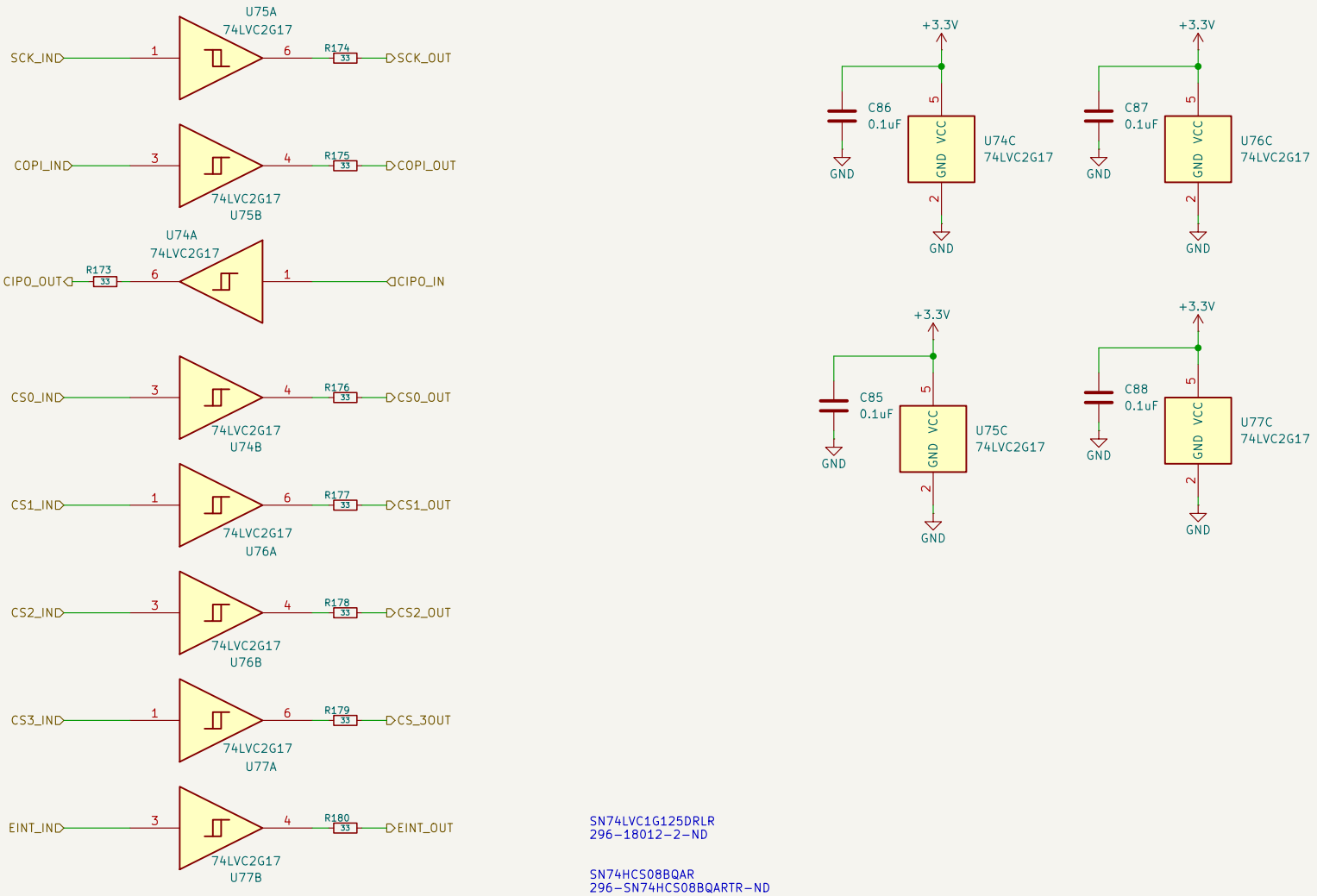
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 46/67



Reiser Lab @ Janelia | HHMI		
Sheet: /Panels/Panel Column P10/		
File: panel_column.kicad_sch		
Title: G6 Teensy 10–14 Arena		
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 47/67



SN74LVC1G125DRLR
296-18012-2-ND

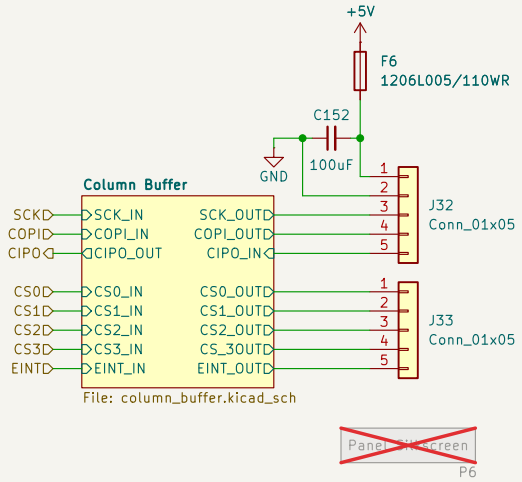
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

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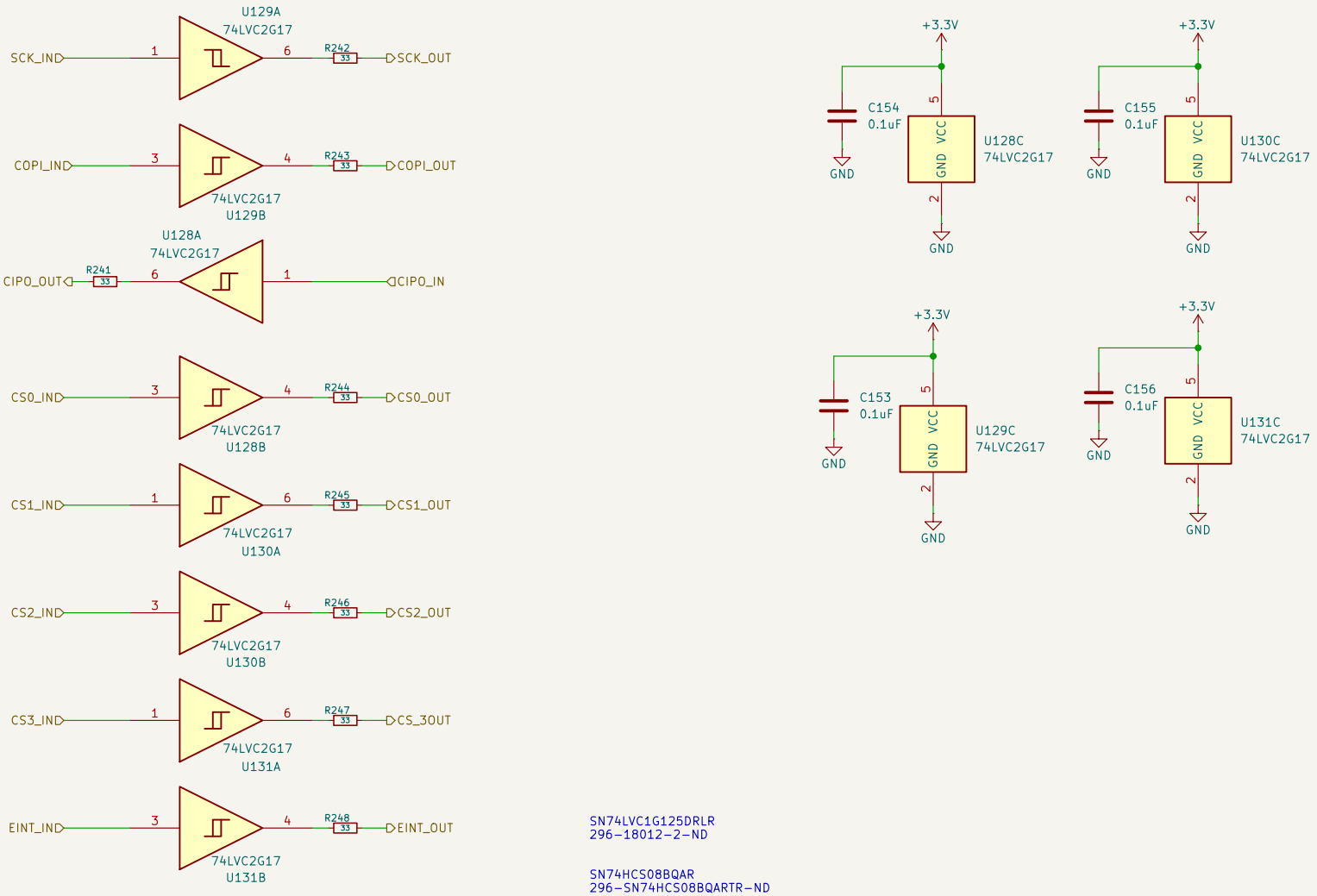
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File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 48/67



Reiser Lab @ Janelia | HHMI		
Sheet: /Panels/Panel Column P6/		
File: panel_column.kicad_sch		
Title: G6 Teensy 10–14 Arena		
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 70/67



SN74LVC1G125DRLR
296-18012-2-ND

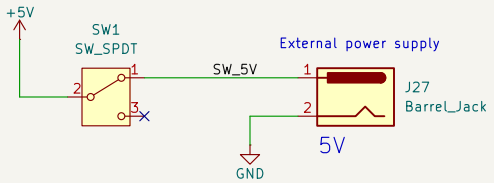
SN74HCS08BQAR
296-SN74HCS08BQARTR-ND

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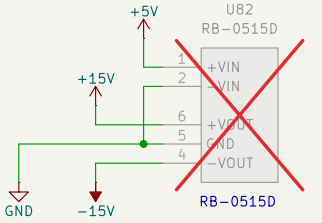
Sheet: /Panels/Panel Column P6/Column Buffer/
File: column_buffer.kicad_sch

Title: G6 Teensy 10-14 Arena

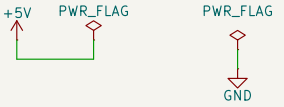
Size: USLetter	Date: 2026-08-03	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 71/67

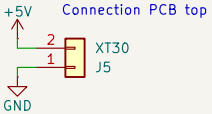


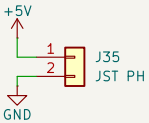
Change DC Jack to
PN DC-053D-20A Korean Hroparts Elec
LCSC C707274



Power rails for
bipolar analog inputs





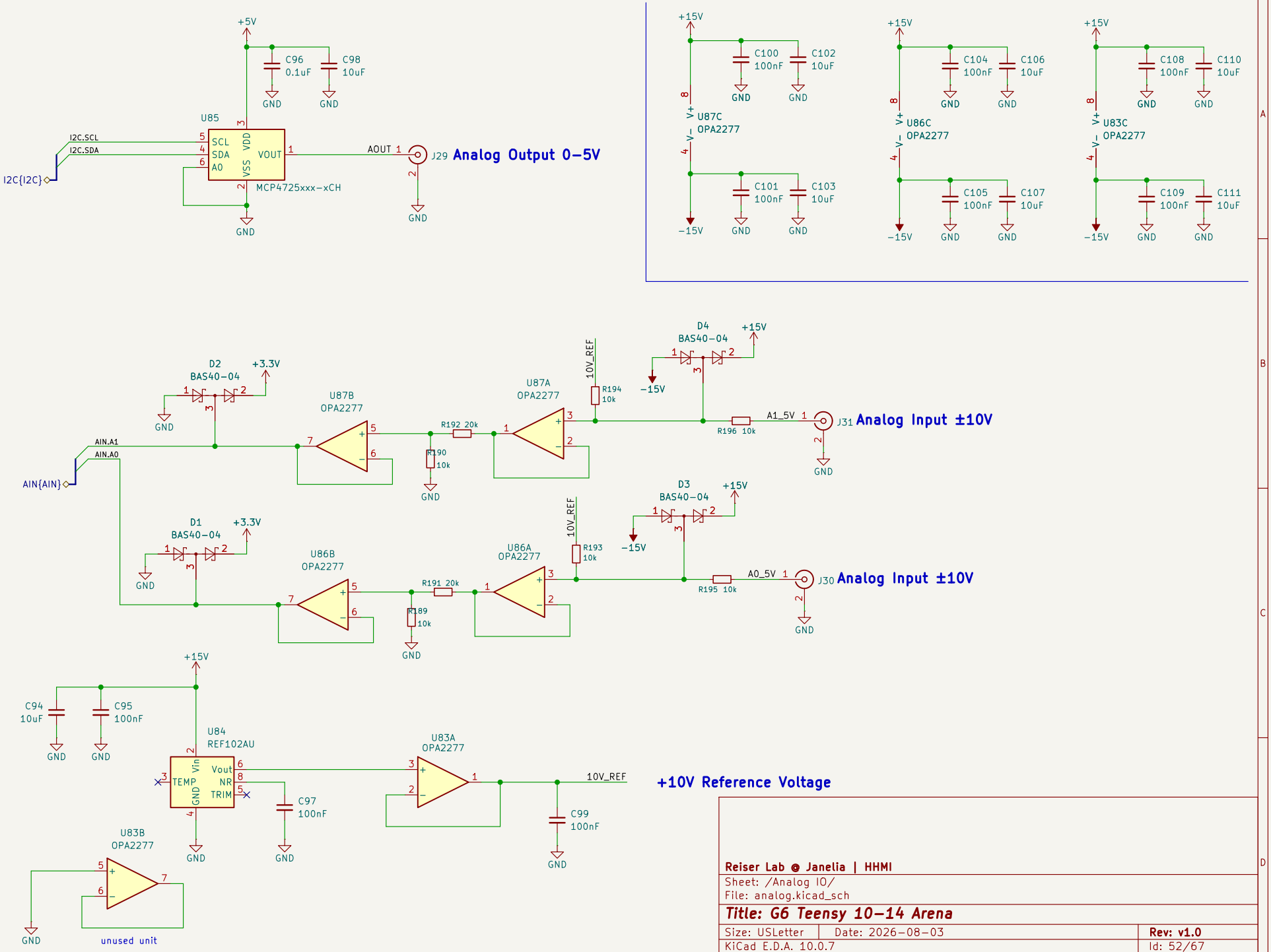


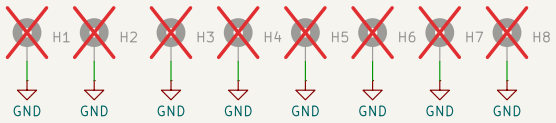
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Sheet: /Power/
File: power.kicad_sch

Title: G6 Teensy 10-14 Arena

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+5V
C96 0.1uF
C98 10uF
GND
U85
I2C.SCL
I2C.SDA
I2C{I2C}
SCL
SDA
A0
VDD
VOUT
VSS
MCP4725xxx-xCH
AOUT
J29
Analog Output 0–5V
+15V
C100 100nF
C102 10uF
U87C
OPA2277
C101 100nF
C103 10uF
−15V
C104 100nF
C106 10uF
U86C
C105 100nF
C107 10uF
C108 100nF
C110 10uF
U83C
C109 100nF
C111 10uF
D2
BAS40-04
+3.3V
U87B
OPA2277
AIN.A1
AIN.A0
AIN{AIN}
R192 20k
R190 10k
U87A
10V_REF
R194 10k
D4
R196 10k
A1_5V
J31
Analog Input ±10V
D1
U86B
R191 20k
R189 10k
U86A
R193 10k
D3
R195 10k
A0_5V
J30
Analog Input ±10V
C94 10uF
C95 100nF
U84
REF102AU
Vin
Vout
NR
TRIM
TEMP
GND
C97 100nF
U83A
C99 100nF
+10V Reference Voltage
U83B
unused unit
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Sheet: /Analog IO/
File: analog.kicad_sch
Title: G6 Teensy 10–14 Arena
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Sheet: /Mount Holes/

File: holes.kicad_sch

Title: G6 Teensy 10-14 Arena

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